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ORIGINAL PAPER

Micro-mechanical Properties of Main Rock-Forming Minerals in Granite Under Microwave Irradiation

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Abstract

Currently, the large-scale crushing technology for high-strength hard rock is struggling to meet the increasing construction demand for deep underground engineering. Although microwave-coupled mechanical rock breaking is safe, clean, and efficient, the weakening mechanism of the mechanical properties of microwave-treated hard rock is unclear, especially for quantitative description in mineralogical terms. Thus, the variation laws of micro-mechanical properties and surface morphology of samples in different parts of ten major granite-forming minerals were systematically studied after being irradiated with 2 kW microwaves for 3 min. Results show that minerals can be classified into three categories based on their micro-mechanical properties: high-hardness ($H > 11$ GPa, quartz and amphibole) and medium-hardness minerals ($6 < H < 11$ GPa, feldspar minerals) with good elasticity ($EER \geq 0.59$) and brittleness ($H/Kc \geq 1.65$), and low-hardness minerals (< 6 GPa, muscovite, phengite and chlorite) with excellent plasticity ($EER \leq 0.46$) and ductility ($H/Kc \leq 0.96$), excluding diopside. After microwave irradiation, chlorite, altered plagioclase, and amphibole were categorized as microwave-sensitive minerals (heating rates above 2.5 °C/s). Among them, amphibole decreased its mechanical parameters by up to 32.7%, while chlorite and altered plagioclase were enhanced. The heating rates of the other minerals are lower than 2 °C/s, and the surface deterioration is not obvious. Peculiarly, the change rates of mechanical parameters of muscovite and phengite are up to 42.3%, which are clearly distinguished from the lower mechanical parameters change rates of other minerals. Clearly, minerals' microwave sensitivity cannot determine the variation rule of their mechanical properties. Differences in indentation morphology depend mainly on minerals' fracture toughness, elasticity, and brittleness parameters. This study reveals the weakening mechanism of the mechanical behavior in granite after microwave irradiation mineralogically, and offers reliable basic data for cross-scale modeling of granite.

Highlights

- Studied micromechanical properties of 10 key granite minerals pre- and post-microwave exposure.
- Explored link between micromechanics and surface indentation morphology in minerals.
- Examined crystal structure's impact on micromechanical features in minerals.
- Tested melt fractions and their micromechanical properties in amphibole and chlorite.

Keywords Microwave irradiation · Granite · Rock-forming minerals · Nanoindentation · Micro-mechanical parameters

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1 Introduction

Confronted with the gradual depletion of shallow resources, the exploitation of deep resources has emerged as one of the greatest challenges of the twenty-first century (Gao et al. 2020; He et al. 2024; Xue et al. 2023; Xie 2019). This emergency forces human engineering activities to delve deeper underground, such as the world's deepest underground laboratory in Jinping, China, which has reached a depth of 2525 m (Zhang et al. 2011). However, the scientific laws and techniques of shallow engineering are unable to solve the difficulties of deep engineering, leading to the blindness and inefficiency of deep engineering activities (Li et al. 2023; Xie et al. 2023).

One of the difficulties is that rock strength tends to increase non-linearly with depth (Xie et al. 2021). Undoubtedly, blasting and mechanical methods are the most commonly applied techniques in rock breaking (Gao et al. 2023; Hassani et al. 2016; Saiang and Nordlund 2009). However, the blasting method brings severe noise and environmental pollution while causing strong disturbance to the surrounding rock mass (Natanzi and Laefer 2014). On the other hand, low cutting rates and severe tool wear are important problems faced in deep hard rock excavation (Kahraman et al. 2020b). According to incomplete statistics, the cost of tool consumption and the time required for maintenance and replacement account for 30–40% or more of the project cost and construction cycle, respectively (Deliormanli 2012; Gong and Zhao 2009; Hassanpour et al. 2011; Xia et al. 2012). Hence, there is an urgent need for methods to achieve safe and efficient crushing of deep hard rock.

Microwave rock breaking has been shown to weaken rock by virtue of its selective heating, high heating rate, instantaneous electronic control, safety, and cleanliness (Wei et al. 2019). In addition, numerous scholars have demonstrated the effectiveness of microwave irradiation in reducing the uniaxial compressive strength, Brazilian splitting strength, and point load strength of rocks (Kahraman et al. 2020a; Lu et al. 2019; Peinsitt et al. 2010; Yang et al. 2022). However, it is important to note that breaking rocks solely using microwave technology requires a significant amount of energy (Maurer 1968). Recognizing this, Lindroth et al. (1993) introduced the concept of "pre-treatment" of rocks before mechanical rock breaking. Hassani and Nekoovaght (2011) presented and analyzed the potential applications of microwaves in hard rock drilling, cutting, and breaking. They further noted that by combining microwave energy with one of the available excavators, such as a roadheader, tunnel boring machine (TBM), or oscillating disc cutter (ODC), the total energy exerted on the rock can be decreased. Simultaneously, it can effectively reduce wear and tear and the frequency of tool changes during the mechanical rock breaking process, thus reducing construction costs and increasing construction efficiency (Hassani et al. 2016; Lu et al. 2020a).

Igneous rocks are the most widespread in the lithosphere, accounting for about 66% of the entire volume of rocks (Bell 2004). They are also the most popular hard rocks found in tunneling and chamber excavations (Zheng et al. 2020), including granite, gabbro, and others. Among these, granite has become the most widely used natural building material due to its dense structure, high strength, and resistance to weathering. However, these characteristics also create difficulties in the construction of underground engineering. For instance, the hard granite strata of the Suai Tunnel increased the vibration amplitude of the shield machine cutter (Yang et al. 2020b). Similarly, the high-strength granite strata caused tool wear and even back-torsion distances to the shield machine during the construction of Line 10 of the Shenzhen Metro in China (He et al. 2020). Therefore, systematic studies are desperately needed to understand granite weakening mechanisms under microwave irradiation to achieve better results in microwave-assisted granite fracturing.

Numerous authors have used granite as an example and studied its heating properties and strength folding patterns when exposed to microwave radiation. Zeng et al. (2019) discovered variations in the causes of granite strength reduction at different temperatures. At low temperatures (300–600 °C), the decrease is attributed to water emission, quartz transformation, and fracture, while at high temperatures (600–800 °C), it is due to localized melting. Qin et al. (2015) exposed standard-sized granite specimens to microwave fields with variable power densities and irradiated them for different durations. The rock strength decreased linearly with increasing microwave irradiation time and power density. Therefore, the researchers concluded that microwave radiation effectively reduced the strength of the rock mass. Hartlieb and Grafe (2017) investigated the effect of microwave radiation on the force of a conical pick when cutting Neuhauser granite with a size of $50 \times 50 \times 30$ cm^3 in a linear cutting test rig. They discovered that the cutting force decreased by about 10% after 30 s of microwave irradiation with 24 kW power using a checkerboard pattern. It was further derived that modifying the irradiation time, irradiation point spacing and frequency can notably improve the efficiency of microwave treatment ahead of mechanical excavation.

Following numerous studies confirming the feasibility and high efficiency of microwave-assisted crushing of granite, several scholars have begun to address the variability of rock deterioration behavior due to differences in mineral types and contents under microwave action. They aim

to explore the mechanisms by which microwaves weaken granite. This is attributed to the fact that rock material is a composite material composed of multiple minerals, making it a typical non-homogeneous material (Viktorov et al. 2014). For example, data from another study by Hartlieb et al. (2012) demonstrate that rock-forming minerals have an impact on the structural and thermal behavior of selected hard rocks. In addition, the study shows that selective heating by microwave radiation is possible when the rock consists of different minerals (absorbers) with varying dielectric properties. Kahraman et al. (2020a) examined granite samples with varying mineral (absorber) contents. The UCS (uniaxial compressive strength) values generally decreased at surface temperatures higher than 200 °C. However, for orthoclase and gabbro, a significant loss of strength was generally observed below 100 °C, which can be attributed to the apparent differences in mineral type and content of the different specimens. Lu et al. (2020a) conducted a study on the impact of microwave heating on three common types of rocks. They demonstrated that rock fracture caused by microwaves takes place when two specific types of minerals are present: those that are sensitive to microwaves and those with high coefficients of thermal expansion. Essentially, disregarding the objective spatial inhomogeneous distribution of the microwave field, the weakening of rocks by microwaves can be attributed to the variability of the dielectric and thermal–mechanical properties of the minerals within the rock (Fan et al. 2018; Zhao et al. 2020; Zheng et al. 2020). Thermal gradients introduced into the rock generate thermal stresses between different minerals (Hartlieb et al. 2017; Toifl et al. 2016), causing pore, intergranular, and transgranular cracks, and even flaking (Kahraman et al. 2020a; Kingman and Rowson 1998; Zheng et al. 2017), which can lead to the rupture of the granite.

Consequently, there has been a substantial body of research into the microwave sensitivity of major rock-forming minerals under microwave irradiation, to estimate the heating behavior of rocks by microwave radiation. Chen et al. (1984) assessed the microwave absorptivity of 40 minerals and found that most silicates, carbonates, and sulfides, some oxides, and some sulfides were transparent to microwave energy, but other oxides were readily heated. Studies at that time concentrated on the mineralization of ores, while basal orogenic (veinstone) minerals were broadly considered to be microwave-permeable minerals. In addition, analogous conclusions were reported by Walkiewicz et al. (1988) and McGill et al. (1988). A shadow has been cast over the future of microwave-assisted rock fragmentation, as researchers argue that microwaves will not be effective in breaking up hard rock. Harrison (1997) contends that this chemical classification is not applicable to all minerals. A counterexample is given: most sulfides have a relatively benign thermal effect, but sphalerite does not. On the other hand, he proposed an analogy between microwaves and light. Dark minerals that absorb light have excellent microwave absorption, while bright minerals that reflect or refract light have inferior microwave absorption. This view is also supported by Ford and Pei. (1967). Zhao et al. (2020) further supplemented this view by demonstrating that the sensitivity of homologous minerals largely depends on their crystal structure. In addition, for minerals of the same group and crystal structure, the sensitivity is determined by their Fe content.

Facing increasingly complex classification conditions, there is no doubt that the dielectric properties ($\varepsilon = \varepsilon' + \varepsilon'' j$, where the dielectric constant ε' determines the amount of energy that can be stored in the material, and the loss factor ε'' determines how much energy can be dissipated as heat (Zheng and Sun 2021)) of rock-forming minerals, inherent to different minerals, are also critical parameters in microwave-assisted rock breaking. These properties play an essential role in the heat and damage generated after microwave radiation (Li et al. 2020; Zheng et al. 2005). Zheng et al. (2020) conducted a test on the dielectric constants and loss factors of 14 commonly found hard rock minerals. They categorized these minerals into three groups: low-loss, medium-loss, and high-loss minerals, based on the magnitude of the loss factor. Meanwhile, in response to the claim that iron content significantly affects the microwave sensitivity of minerals (Chen et al. 1984; Lu et al. 2017), the authors state that no significant dependence of the loss factor on any chemical element or chemical element combination has been found. They also provide the counter-examples that olivine, which has the highest iron content, actually has the lowest loss factor. On the other hand, the authors showed that in the case of crystalline systems, the loss factor presents a random result. That is, there is no specific relationship between the loss factor and the crystal system, a view shared by Harrison (1997).

The most commonly adopted techniques in testing the dielectric properties of rock-forming minerals are the resonant cavity perturbation technique and the short-circuit waveguide method, with the former technique being more suitable and versatile (Chen et al. 1999; Kraszewski and Nelson 1994; Ulaby et al. 1990). However, this testing technique requires specialized network analyzers and customized resonant cavity waveguides (Zheng and Sun 2021), which are not accessible to researchers. More notably, the standard deviation of the loss factor of the high-loss ($\varepsilon'' \geq 1$) minerals measured by Zheng et al. (2020) is relatively high. This is because the principle of resonant cavity perturbation technique cannot maintain the sound when the loss factor is high. Consequently, more scholars have used the method of heating rate testing to quantify the microwave absorption capacity of minerals (Liu 2018; Lu et al. 2017; Tian et al. 2019; Zhao et al. 2020; Zheng and Sun 2021). This indirect method heats the same mass or volume of minerals in microwave

equipment under identical heating conditions and obtains the heating rate. This provides the most direct concept of the temperature increase due to microwave absorption. The heating rate of minerals is related to their density, specific heat, and loss factor. The differences in density and specific heat between various types of minerals are small, so the heating rate effectively reflects the dielectric loss factor (ε") of minerals (Liu 2018). However, it should be emphasized that the heating rate of minerals is strongly influenced by the provided microwave parameters, the size of the microwave cavity, and the electromagnetic impedance matching of the microwave source and load. The results of testing using different systems may be dissimilar or even contradictory (Kingman and Rowson 1998). Therefore, these tests are exploratory and descriptive in nature. In addition, the heating rate of minerals is not an intrinsic parameter and is unlikely to apply to multi-physics field simulations (Zheng et al. 2020). What is most interesting is that when Nicco et al. (2020) investigated the link between crack extension and mineralogical control in granite under microwave irradiation, they found that a strong microwave absorber was not a necessary condition for cracking. They also found that the microwave response could not be predicted based on dielectric constant, color, water, or the iron content of the constituent minerals. For instance, biotite (dark brown to black) and amphibole (black), both dark minerals, contain bound water and exhibit high dielectric constants. Yet, under the same conditions, biotite does not crack, whereas amphibole is disproportionately associated with cracking. This further illustrates that the cracking and damage of hard rock cannot be accurately predicted solely by the thermo-physical and electrical properties of rock-forming minerals. Therefore, more attention should be paid to the changing pattern of the mechanical properties of the minerals after microwave action, which is closely linked to the macroscopic deformation and injury characteristics of non-homogeneous materials like rocks (Li et al. 2019c; Zhang et al. 2021).

It is difficult to prepare standard specimens for some rock-forming minerals, such as mica minerals with layered cleavage and chlorite minerals. Even in the case of complete, large-sized minerals, the varying content of impurities in the minerals leads to deviations in the mechanical test results from the mechanical properties of the minerals themselves. For this reason, the traditional macro-mechanical test is not applicable to rock-forming minerals. The nanoindentation testing technique can compensate for the limitations of traditional testing methods. By employing the nanoindentation micromechanical testing technique, it is possible to acquire load–depth curves (Ma et al. 2022; Tang et al. 2022) for various rock-forming minerals. In addition, valuable data such as hardness and Young's modulus (Liu et al. 2023; Maruvanchery and Kim 2020), fracture toughness (Ante et al. 2018; Liu et al. 2023), elastic–plastic deformation work (Liu et al. 2022; Lu et al. 2022), residual stresses (Suresh and Giannakopoulos 1998; Xu and Li 2008), brittleness (Fan et al. 2019; Lawn and Marshall 1979), and creep behavior (Espinoza et al. 2022; Sharma et al. 2019) can be inferred and calculated from this technique. These properties contribute to the development of predictive petrophysical models (Li et al. 2019a), the quantification of elastic and plastic deformation resistance (Joslin and Oliver 1990), the analysis of deformation and fracture relative sensitivities (Lawn and Marshall 1979) (i.e., brittleness and toughness transition laws), and the understanding of fracture initiation and extension processes (Bhushan 2017; Gupta et al. 2018). Although macro-mechanical tests can also yield relevant mechanical parameters, the relationship between micro-mechanical properties and crack development is closer. Meanwhile, nanoindentation testing technique has greater advantages in studying the micro-mechanical properties of minerals. In addition to the advantages of high resolution of load and displacement, easy sample preparation, short time-consuming and less destructive creep tests, and its own self-contained property summarized by Ma et al. (2020), nanoindentation testing technique also possesses a relatively low dispersion of experimental results (Wang et al. 2023).

Another interesting point is that numerous scholars have confirmed, through various methods, that the internal temperature of rocks heated in a microwave is higher than their surface temperature. For instance, Peinsitt et al. (2010) conducted a study where they drilled holes along the cylindrical axis of a basalt sample and inserted thermocouples to measure internal temperatures. The results revealed that after 260 s of irradiation at 3 kW microwave power, the internal and surface temperatures of the basalt samples reached 1220 °C and 580 °C, respectively. Sun and Ma (2022) and Lu et al. (2020a) found a sharp increase in the temperature of the infrared camera after the specimen was broken or melted. Using numerical methods, Liang et al. (2023) found that under 2 kW microwave irradiation, some "steep hot spots" were first formed inside the specimen. Subsequently, the range of hot spots gradually grew and the temperature gradient gradually decreased, indicating that the high-temperature zone of microwave heating was first formed in the interior. However, the final purpose of the microwave irradiation experiment is not to gain the internal temperature of rocks, but to acquire the mechanical properties inside the samples. Numerical methods can achieve this, for example, the internal temperature and stress–strain distribution characteristics of single-crystal minerals have been investigated using the finite difference method and finite element method, respectively (Jones et al. 2005; Wang et al. 2008; Wang and Djordjevic 2014). Nevertheless, some assumptions made during the development of these models can limit the reproducibility of simulation results in experiments. For instance, neglecting heat damage in electromagnetic–thermal conversion, the influence of water within the sample (Liang

et al. 2023), and disregarding the impact of temperature gradients during the cooling process, as well as changes in thermal and mechanical properties after temperature homogenization (Wang and Konietzky 2019). Therefore, it has been a challenge to accurately detect mechanical variations within rocks under microwave action. The nanoindentation test technique can be utilized to examine the micro-mechanical parameters of various sections of rocks after microwave action, thanks to its small sample specification requirements and simple sample preparation procedure. This enables us to quantify the weakening patterns of micro-mechanical properties in different portions of specimens following microwave exposure.

In this study, the nanoindentation test technique is introduced to systematically investigate the variation rules of micro-mechanical properties and morphological features in different parts of the granite's major rock-forming minerals before and after microwave action. Specifically, ten minerals were classified as high-hardness (Di, Q, and Am), medium-hardness (Kf, Ab, Pl, and Apl), and low-hardness minerals (Ms, Phn, and Ch) based on a range of micro-mechanical indicators from the nanoindentation tests. In addition, all high- and medium-hardness minerals except Di have excellent elasticity and brittleness, while all low-hardness minerals have excellent plasticity and ductility. After microwave irradiation at 2 kW power for 3 min, combined with surface temperature monitoring, thin-section observation, XRD (X-ray diffraction) analysis, and statistical analysis to compare the micro-mechanical properties of minerals, it was found that the sample site, crystal structure, impurities and temperature all have an impact on the results of micro-mechanical testing of the minerals, not only related to their heating rate. Therefore, the classification of microwave sensitivity of minerals in the existing studies cannot truly reflect the change pattern of the minerals' mechanical properties. In addition, the melt reaction mechanisms of Ch and Am were analyzed using XRD analysis and K-mean clustering calculations, and the melt fractions were classified into three categories based on their strengths. In addition, the surface indentation morphology of different minerals is mainly controlled by the fracture toughness, elasticity, and brittleness parameters of the minerals. This study is conducive to exploring the weakening mechanism of microwave-treated granite, and providing a new method for large-scale traditional mechanical tests. It also serves as an experimental basis for the industrial application of microwave-assisted crushing in hard rock.

2 Experimental Sample and Protocol

2.1 Sample Preparation and Component Test

As summarized by Xu and Qiu (2010) and Tarbuck et al. (2005), granites are mainly composed of rock-forming minerals including quartz, mica, plagioclase feldspar, sodium feldspar, potassium feldspar, hornblende, and pyroxene. Accordingly, the raw ores collected in this research consist of quartz (Q), albite (Ab), K-feldspar (Kf), plagioclase (Pl), altered plagioclase (Apl), muscovite (Ms), phengite (Phn), amphibole (Am), diopside (Di), and chlorite (Ch). Actually, chlorite is a clay mineral belonging to metamorphic rocks, yet some minerals (i.e., pyroxene, amphibole, and biotite) are often altered to chlorite in igneous rocks. For instance, chlorite minerals are contained in pegmatites of Li et al. (2019b), granodiorites of Nicco et al. (2020), and granites of Kahraman et al. (2020b). Consequently, chlorite was chosen as one of the subjects for this investigation.

In previous studies, some scholars (Liu 2018; Walkiewicz et al. 1988; Zhao et al. 2020) crushed minerals into powders, weighed an equal mass (20–40 g) or volume of mineral powders, and tested them at a fixed microwave power (around 2 kW) and time (3–5 min). The advantage of this research method is that it allows for easy control of mineral quality consistency and ensures mineral purity. However, the disadvantage is that it disrupts the cleavage of the minerals (Zheng et al. 2020). Concurrently, the results of Liu (2018) and Lin et al. (2023) revealed that when sample bulk and powder with the same mass or volume were heated under identical conditions, the rupture phenomenon and heating rate of the bulk and powder differed considerably, which was related to the microwave action mechanism of the samples. On the other hand, some scholars (Lu et al. 2017; Tian et al. 2019) have used irregular lumpy minerals as the subject of their research. These minerals preserve the cleavage structure, but they do not ensure consistency in terms of mass or volume and may contain impurities to varying degrees. After comparing the advantages and disadvantages of both types of samples, this paper has decided to use intact large raw minerals processed into 50 × 50 mm cylinders as the subject of the microwave irradiation experiment. This guarantees the same volume of rock-forming minerals as well as the integrity of the deconvolution and grain structure, while avoiding shape and size effects on microwave-heated minerals. Such a concept of processing large-size minerals as cylinders was first proposed by Zheng et al. (2020).

Nonetheless, there are two key issues encountered in the actual processing: one is the irregular shape of large-sized minerals, and the fact that layered silicate minerals are prone to misalignment and slagging along the cleavage, resulting in a low yield. Considering that no conventional macro-mechanical experimental studies would be carried out, it was ensured that only three pieces of each mineral sample were processed, which were close to each other in terms of mass or density (samples 4–6). Another point is that large-sized minerals contain unequal amounts of impurities. Therefore, a combination of mineral thin-section testing and XRD semi-quantitative analysis was used to further

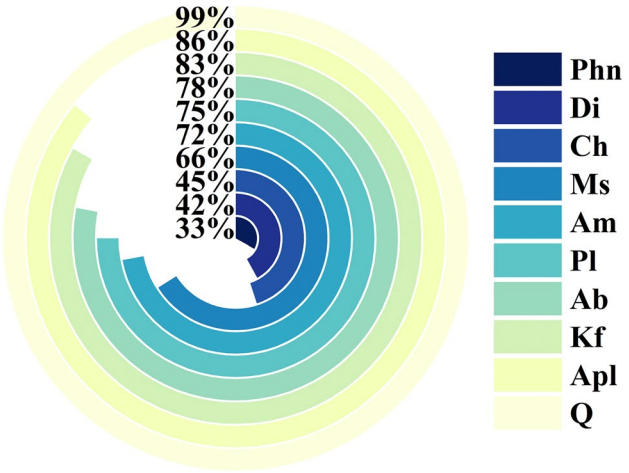


Fig. 1 Mineral purity of the original ores of the ten rock-forming minerals

analyze the effects of the type and content of impurities and microstructure on the heating efficiency of the minerals. The purity of 10 rock-forming minerals is shown in Fig. 1, and their impurity types and corresponding contents are shown in Table 1. Apparently, most of the minerals have a purity of about 70% or more, except for Phn, Di, and Ch, which contain relatively high levels of impurities. Among these, the main impurity in Phn is quartz (66%), the main impurity in diopside is dolomite (43%), and the main impurity in chlorite is plagioclase (46%). It is important to note that quartz and feldspar minerals are the main impurities in these rock-forming minerals. The impact of impurities on the heating properties of rock-forming minerals will be discussed in the relevant sections.

As the nanoindentation specimens have no strict dimensional demands, the relevant samples were selected from the corresponding positions of the crushed cylindrical mineral samples, and it was also ensured that the thickness of these samples was greater than 10 times h_{max} (1000 nm) (Randall et al. 2009). As shown in Fig. 2, each sample was placed into a cylindrical abrasive with an inner diameter of 30 mm and cured in epoxy resin. The synthetic samples were cylinders with a diameter of 30 mm and a height of 10–15 mm. Next, rough grinding was carried out using P320, P600, and P1200 sandpaper for more than 30 min per grit. Then, a Struers polishing machine was used to progressively polish the samples to a finer particle size of up to 0.05 µm. The samples were also ultrasonically cleaned with anhydrous ethanol throughout the polishing process. Lastly, the samples were placed in an oven at 110 °C and dried for 24 h. The three types of samples (untreated, surface, and interior) are shown in Fig. 2.

2.2 Experimental Equipment and Method

Peinsitt et al. (2010) have confirmed that the presence of water in minerals and rocks greatly affects their heating properties. Therefore, all mineral species in this study were subjected to a drying process in an oven at 110 °C for 48 h before conducting microwave irradiation tests. This allows the adherent water adsorbed on the surface of the mineral particles and the interlayer water present between the structural layers of the layered silicate mineral lattice to be completely released, avoiding any influence on the microwave tests. Subsequently, the microwave-irradiated mineral test was conducted using the deep hard rock microwave fracturing equipment independently developed by the State Key Laboratory of Intelligent Construction and Healthy Operation and Maintenance of Deep Underground Engineering at Sichuan University. The device consists of a 15 kW adjustable microwave transmitter, BJ26 waveguide circulator, BJ26 rectangular waveguide (86.36 × 43.18 mm), BJ26 triple-pin power modulator, water load, and an LS-C heating chamber (500 × 500 × 300 mm). The system operates at a frequency of 2.45 GHz. Referring to the microwave heating conditions of Lu et al. (2017), whose study also focused on mineral masses, ten granitic rock-forming minerals (three samples of each mineral were set up for parallel testing)

Table 1 Impurity content of the original ore of the ten rock-forming minerals

Mineral type	Impurity a (Content/%)	Impurity b (Content/%)	Impurity c (Content/%)	Impurity d (Content/%)	Impurity e (Content/%)
Kf	Quartz (16%)	Biotite (1%)	–	–	–
Ms	Quartz (37%)	Magnesioferrite (1%)	–	–	–
Phn	Quartz (66%)	Brucite (1%)	–	–	–
Di	Dolomite (43%)	Calcite (7%)	Tremolite (5%)	Muscovite (2%)	Quartz (1%)
Ch	Plagioclase (46%)	Albite (5%)	Quartz (3%)	Epidote (1%)	–
Ab	Quartz (18%)	Muscovite (3%)	Ercitite (1%)	–	–
Q	Cuprite (1%)	–	–	–	–
Pl	Quartz (20%)	Microcline (4%)	Biotite (1%)	–	–
Apl	Quartz (7%)	Muscovite (5%)	Albite (2%)	–	–
Am	Plagioclase (20%)	Biotite (4%)	Quartz (2%)	Carnegieite (1%)	Sphene (1%)

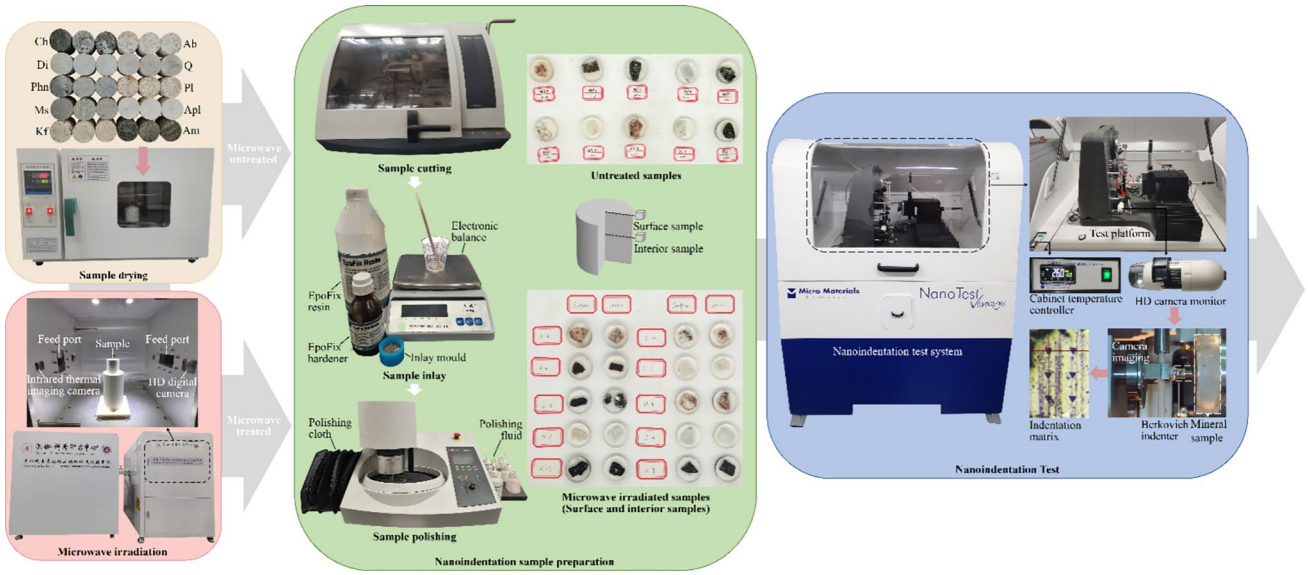


Fig. 2 Experimental procedure

were irradiated by microwave radiation of 2 kW power for 3 min. Meanwhile, the surface temperature and overall cracking of the minerals were captured in real time using the built-in infrared thermal camera (temperature measuring range: − 20–900 °C; optical resolution: 382 × 288 pixels) and high-definition digital camera of the equipment. The specific testing procedure is shown in Fig. 2.

In Fig. 2, the nanoindentation experiment is based on Micro Materials' nanoindentation test system, combined with its own positioning optical microscope, and a Berkovich indenter is selected for the test. The displacement resolution was 0.01 nm and the load resolution was 50 nN. The selection of indentation depth has an impact on testing the mechanical properties of materials (Bennett et al. 2015; CaÅa et al. 2017; Li et al. 2019c). For example, Lei et al. (2021a) observed that the hardness and elastic modulus of materials were larger when the indentation depth was low and gradually stabilized when the depth was increased to 350 nm. On the other hand, excessive indentation depths can cause the measured values to be affected by the minerals beneath or adjacent to them, i.e., the surround effect and the basal effect (Lu et al. 2020b; Wu et al. 2020; Yang et al. 2020a). Therefore, the maximum indentation depth was set to 1000 nm for this study. In addition, the effect of roughness can be minimized by selecting a larger indentation depth (Lucca et al. 2010) and a smoother indentation area (Donnelly et al. 2006). Furthermore, achieving thermal equilibrium prior to indentation is necessary to avoid thermal drift effects. Feng and Ngan (2002) indicated that if the time between the start of testing and the start of unloading is less than $(S \cdot h_c)/P$ (where h_c represents the contact depth, P is the unloading rate, and the loading and unloading rates are 2 mN/s in this paper), the impact of thermal drift on the calculated nanoindentation modulus is negligible. The slower loading and unloading rates, along with the system's automatic calibration procedure, enable minimal effects of thermal drift on the data collected in this study.

Due to the compositional heterogeneity and variability of the surface roughness of the minerals (Mo et al. 2022), three or five sites with flat and smooth surfaces were selected for the nanoindentation test (three sites for untreated specimens and five sites for treated specimens). At each site, arrays of 2 × 3 indentations were conducted. Ulm et al. (2010) demonstrated that the size of the region of interaction between indentations, under the influence of the indenter, can be visualized as a hemisphere with a radius of $4h_{max}$. For cementitious materials, the linear scale of the interaction region is 1 to 3 μm (Trtik et al. 2009). In contrast, mudstone is an extremely soft rock that has a larger interaction region compared to cementitious materials. Therefore, the spacing is generally greater than or equal to 10 μm or more (Bittnar et al. 2009; Wang et al. 2009), allowing the plastic deformation zone to expand uniformly within the material (ISO 14577-1 2003). Thus, for this paper, the indentation distance is taken as 20 μm. Put differently, 18 nanoindentation trials were performed on each untreated specimen, and 30 nanoindentation trials were performed on each treated specimen to ensure consistency and reliability of the measurements. All load–depth curves were recorded. Considering the random distribution of micro-defects in different minerals, which leads to the dispersion of the results, the microwave sensitivity of the micro-mechanical parameters was revealed by statistically analyzing their mean values and distribution characteristics after excluding outliers.

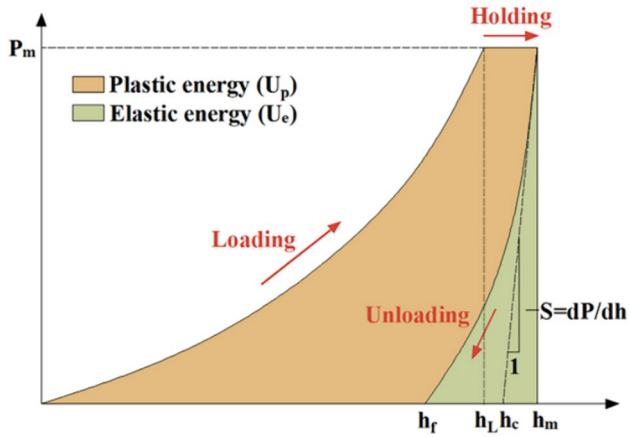


Fig. 3 Typical load–depth curves for nanoindentation

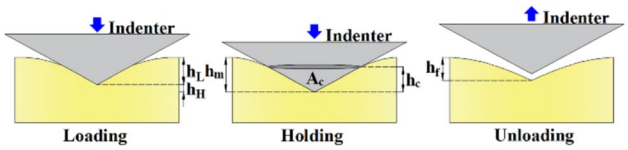


Fig. 4 Indenter conditions and related parameters during loading, holding and unloading periods (h_f, the residual depth; h_L, the load depth; h_H, the creep depth; h_c, the contact depth; h_m, the maximum depth; A_c, the contact area.)

2.3 Nanoindentation Principle and Typical Curve

The nanoindentation test technique belongs to the instrumented indentation test technique in depth sensing technology. This technique employs classical solutions of the contact problem (Hertz 1882; Sneddon 1965; Tabor 1948) to develop precise relationships between the stiffness, deformation, elastic modulus, and elastic recovery of the materials (Abedi et al. 2016; American Society for 2007; Oliver and Pharr 1992, 2004). It has been extensively applied to the characterization of micro-mechanical parameters in various types of rocks (Lu et al. 2022; Tang et al. 2023, 2022; Voltolini et al. 2021).

As shown in Figs. 3 and 4, nanoindentation mainly consists of three stages: loading, holding, and unloading. In the loading stage, the indentation volume deforms elastically when the load applied to the indenter is sufficiently small. As the load increases, the volume with the maximum elastic deformation experiences an elastic–plastic transition, causing plastic deformation. When the load is sufficiently large, the indentation volume enters a fully plastic stage. After reaching the maximum depth, it was held for 10 s to eliminate the strain hysteresis effect that occurs during loading (Miller et al. 2008) and to uncover the deformation mechanism of the indented material over time. In the unloading stage, only elastic deformation can be regained as the indenter is removed, making it possible to determine parameters such as stiffness, hardness, and elastic modulus of the materials.

The Oliver–Pharr method, the Cheng–Cheng method, and the Ma method are three nanoindentation test analysis methods that are recommended in the Chinese standard GB/T 22458-2008 (2008). Specifically, the Oliver–Pharr analytical method employs elastic contact theory to analyze the elastic–plastic deformation problem associated with indentation, and it has undergone extensive experimental validation and refinement. Consequently, it has gained widespread acceptance and is now considered the international standard approach. Moreover, this method is characterized by its simplicity and practicality. Therefore, this study calculates the micro-mechanical parameters of rock-forming minerals based on the Oliver and Pharr method.

The contact stiffness is specified as the slope of the tangent line at the maximum load on the unloading curve and can be expressed as follows:

$$S=\left.\frac{dP}{dh}\right|_{h=h_m}. \tag{1}$$

The reduced modulus is calculated as below:

$$E_r=\frac{\sqrt{\pi}}{2\beta}\cdot\frac{S}{\sqrt{A_c}}. \tag{2}$$

The contact area A_c is related to the contact depth h_c, which can be expressed as follows:

$$A_c=f\left(h_c\right). \tag{3}$$

For the Berkovich indenter, the contact depth can be expressed as follows:

$$h_c=h_m-\varepsilon\frac{P_m}{S}. \tag{4}$$

Hardness is indicated as below:

$$H=\frac{P_m}{A_c}. \tag{5}$$

According to the Hertz contact theory (Hertz 1881), the Young's modulus of the material can be calculated as follows:

$$E=\left(1-\nu^2\right)\left[\frac{1}{E_r}-\frac{1-\nu_i^2}{E_i}\right]^{-1} \tag{6}$$

where β is the indenter correction factor and ε is a constant related to the shape of the indenter. For the Berkovich indenter, $\beta=1.034$ and $\varepsilon=0.75$. As shown in Fig. 4, h_c is the

contact depth. A_c is the contact area which can be expressed as $A_c = 24.56h_c^2$(Oliver and Pharr 1992). E_r is the reduced modulus, which represents the interaction effect between the indenter and the material. E_i and v_i denote Young's modulus and Poisson's ratio of the indenter, respectively. For the diamond indenter used in this study, $E_i = 1141$ GPa and $v_i = 0.07$. v is the Poisson's ratio of the indented material. It can be reasonably assumed based on the relevant literature (Ji et al. 2018; Mavko et al. 2020). This is because a sensitivity analysis of Poisson's ratio with 40% uncertainty shows that the uncertainty in Young's modulus is only 5% (Hay 2009). In this paper, the micro-mechanical properties of 10 rock-forming minerals were determined based on the above testing principles.

3 Experimental Results and Analyses

3.1 Microwave Effects on Surface Morphology

Figure 5 shows the surface optical micrographs and locally enlarged indentation morphology of different parts of various minerals under microwave irradiation. It is observed that after microwave treatment, the surface pores of the interior and surface samples of the minerals are abundant, with locally embedded melt particles and a complex distribution of intra- and inter-crystalline cleavages. In the case of feldspar minerals, the surface morphology changes of Kf, Ab, and Pl are similar, as shown in Fig. 5a, f, and h, with small black pores and a few cracks on the surface of the surface samples. However, more pores, wider cracks, and denser cracks are found on the surface of the interior samples, along with small particles of melt interspersed in the cracks. As can be seen from Fig. 6, Apl possesses a patchy structure, and plagioclase mostly exists in matrix form, with a grain size mostly below 0.2 mm. Therefore, the changes in the surface morphology of samples from different sites of Apl after microwave heating are different from those of other feldspar minerals. As shown in Fig. 5i, the surface of the surface sample showed more small-volume holes and localized large-volume holes with no obvious cracks. In contrast, the interior sample's surface is observed to be more densely populated with holes, especially the number of large-volume holes, but intra- and inter-crystalline cracks are still not obvious. According to the findings presented in Fig. 5g, the surface sample of quartz exhibits minimal closed-state cracks, a slight increase in crack openings in the interior sample. Conversely, the surface sample of diopside, as depicted in Fig. 5d, demonstrates an absence of cracks and only a discrete arrangement of small-volume pores. However, the interior sample of diopside displays a considerably higher number of pores, which are interconnected by cracks. The surface degradation of amphibole is very severe, as shown in Fig. 5j. The surface sample shows a distribution of large-volume holes and small-volume holes along grain boundaries that are partially filled with melt. In contrast, the interior sample exhibits more densely packed large-volume holes that are filled with melt. As shown in Fig. 5b, c, e, Ms, Phn, and Ch are highly susceptible to flaky exfoliation on the sample surface due to their natural laminar cleavage and poor interlayer adhesion. Since quartz particles are also interspersed in the mica minerals, along-crystalline cracks are usually distributed between the mineral particles after microwave heating. On the other hand, through-crystalline cracks are more common in layered silicate minerals, which tend to be produced between the layers. The same conclusion was drawn by Mo et al. (2022). In addition, there are small particles of melt interspersed between the layers, as shown in Figs. 5b4–6 and 5c4–6, which can be seen in the surface morphology when pressed parallel to the disintegration direction.

3.2 Microwave Effects on the Load–Depth Curves

Load–depth curves are the basis for analyzing the deformation behavior and mechanical properties (e.g., H, E) of minerals or mineral interfaces at the micro-scale (Liu et al. 2023). Based on the observation of the morphology, structure and distribution characteristics of different minerals in thin sections, the microscope of the nanoindentation system is used to locate the mineral indentation test sites. Full load–depth curves (18 curves for untreated minerals and 30 curves for microwave-treated minerals) were obtained for all 10 major rock-forming minerals of the granite, as shown in Fig. 7. The known hardness was defined as the maximum load applied perpendicular to each unit of contact area (Fischer-Cripps 2011; Oliver and Pharr 1992), as shown in Eq. (5). After setting the indentation depths to 1000 nm, the indentation areas were almost the same. Therefore, the mineral hardness is only related to the maximum load. Among them, the maximum loads of Di, Q, and Am reached 180 mN, 145 mN, and 150 mN, respectively, implying that all three are hard minerals. This is followed by the four feldspar minerals, with the maximum loads of Kf, Ab, and Pl reaching more than 105 mN, and that of Apl reaching 90 mN. The smallest maximum loads are for Ms, Phn, and Ch, which are all 2:1 type layered silicate minerals. It is noteworthy that the distribution of the load–depth curves is more centralized for the minerals Di, Q, and Am, which have larger maximum loads. Ms, Phn, and Ch have the most dispersed distribution of load–depth curves. This confirms the phenomenon that the dispersion of the curves decreases with increasing maximum load as found by Ma et al. (2022). Meanwhile, the difference in the angle between the indenter press-in direction and the lamination direction is also one of the key reasons for the large dispersion of the curves of

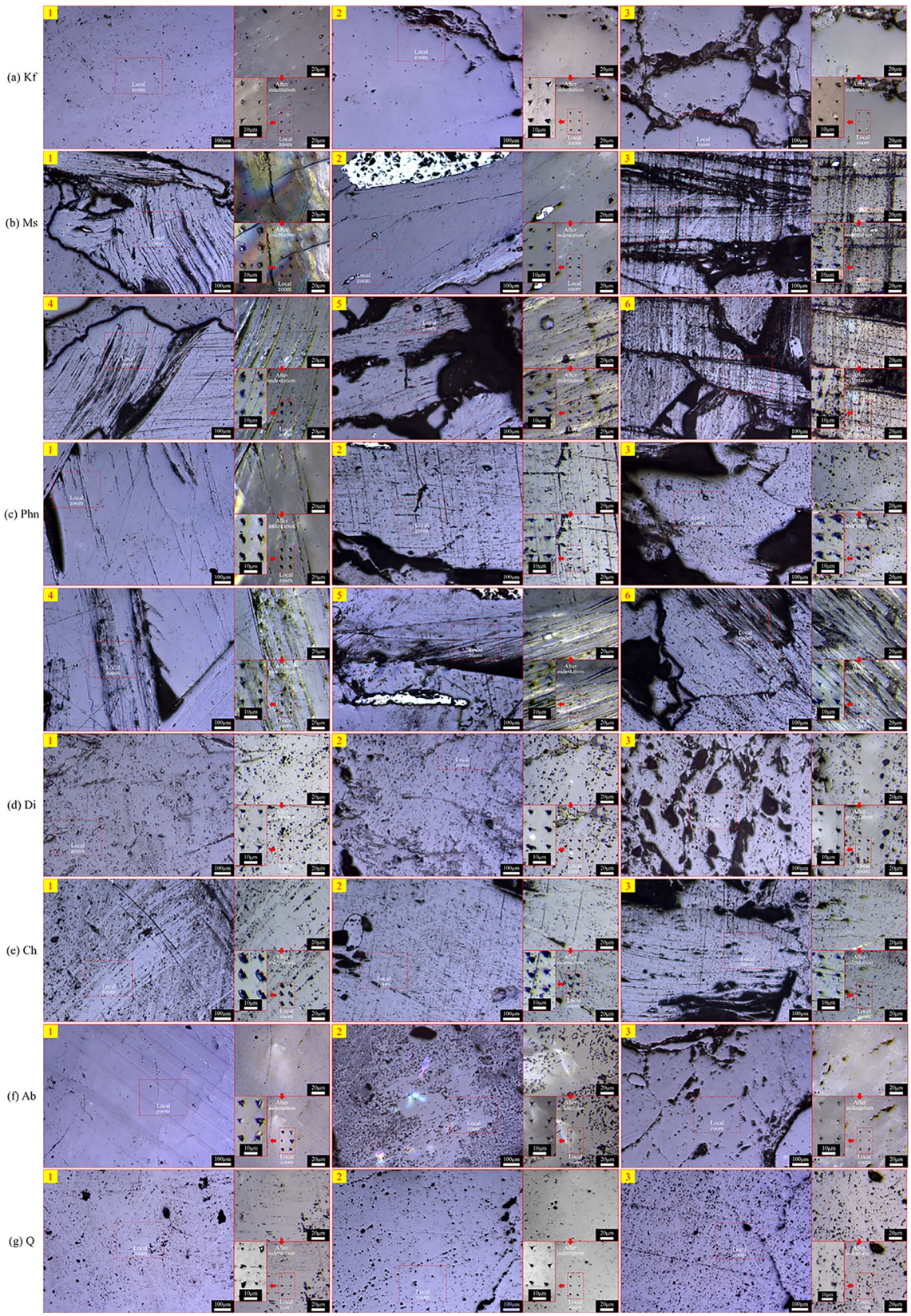


Fig. 5 Surface condition and indentation morphology of different parts of different minerals under microwave irradiation; 1/4—untreated, 2/5—surface, 3/6—interior; 1–3 in (**b**) and (**c**) indicate that the indentation direction is approximately vertical to the layer direction, while 4–6 indicate that the indentation direction is approximately parallel to the layer direction

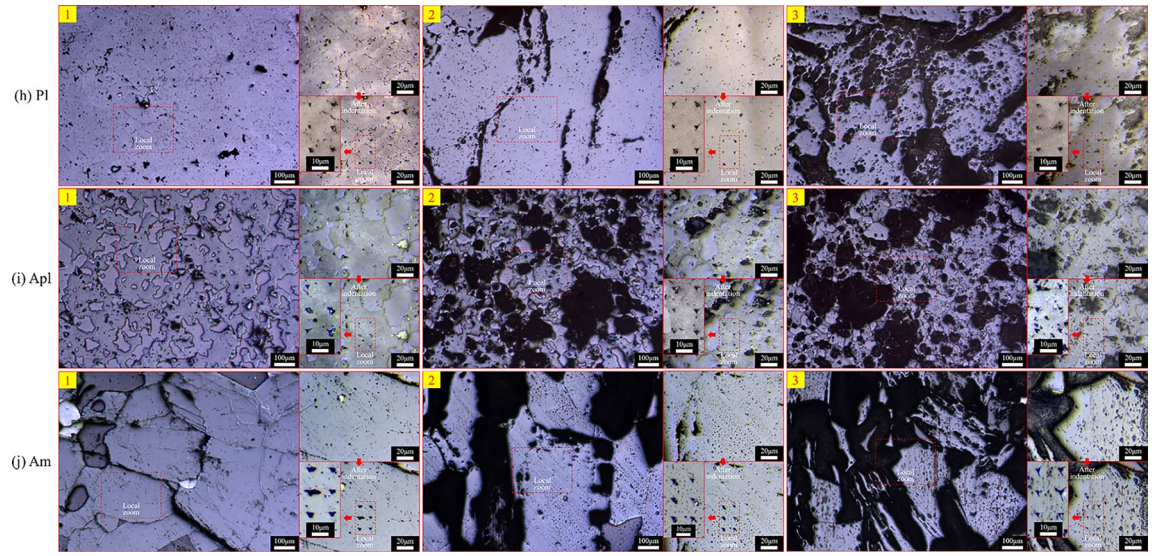


Fig. 5 (continued)

Ms and Phn. As shown in Fig. 7b-1 and c-1, the maximum load of the load–depth curves when indenting perpendicular to the solvation direction of Ms and Phn is larger, with more pop-in events during the indentation phase and smaller residual depths after unloading. Conversely, the mechanical properties obtained when indenting parallel to the solvation direction present the opposite pattern. After the microwave effect, the maximum loads of the minerals showed varying levels of increase or decrease. For example, the maximum loads of Di, Ms, Ab, Q, and Pl showed a small decrease, and that of the minerals' interior samples decreased to a greater extent. The decreases in the maximum load values are slightly greater for Kf, Phn, and Am. The difference is that Phn and Am both have a greater decrease in the maximum load for mineral surface samples. The anomalies are Ch and Apl, both of which show an increase in the mean value of the maximum load. This predicts that the two are strengthened after microwave action. A similar strengthening effect also occurs in the biotite contained in granites after high-temperature action, as tested by Mo et al. (2022).

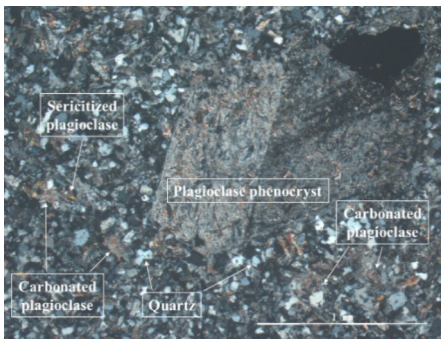


Fig. 6 Microscopic observation of Apl thin section

In addition, it has been noted that the depth of the residual indentation is determined by the hardness, which quantifies the mineral's ability to resist deformation (Lawn et al. 1976). However, this study does not confirm this, as shown in Fig. 7. The residual depths of Di, Q, and Am, which have higher hardnesses, vary considerably, measuring 514.07 nm, 356.51 nm, and 431.21 nm, respectively. Furthermore, the residual depths of the layered silicate minerals are all above 500 nm. As for the feldspar minerals, the residual depths of Ab, Pl, and Apl are slightly different: 387.18 nm, 373.41 nm, and 386.46 nm, respectively. However, the residual depth of Kf is the smallest, measuring only 286.65 nm. It can be observed that minerals with lower hardness exhibit a greater residual depth after unloading. However, the hardness of harder minerals does not show a strict linear correlation with the residual depth. After microwave irradiation, the residual indentation depths of different minerals underwent varying changes. Regarding Q, Ab, Pl, and Apl, a marginal increase in the mean residual indentation depth of the specimens was observed after microwave treatment. Kf displayed a residual depth of approximately 400 nm, which closely resembled the other three feldspars. On the other hand, Ms and Phn experienced a greater increase in residual indentation depths after microwave treatment, reaching a range of 600–700 nm. The residual indentation depth of Ch, which is also a layered silicate mineral, decreased significantly after microwave treatment. The mineral that showed a similar anomaly was Di, but the decrease in the depth of Di was smaller, so it might be a chance phenomenon. The average residual depth of the surface sample of the microwave-treated Am increased significantly, but the average residual depth of the interior

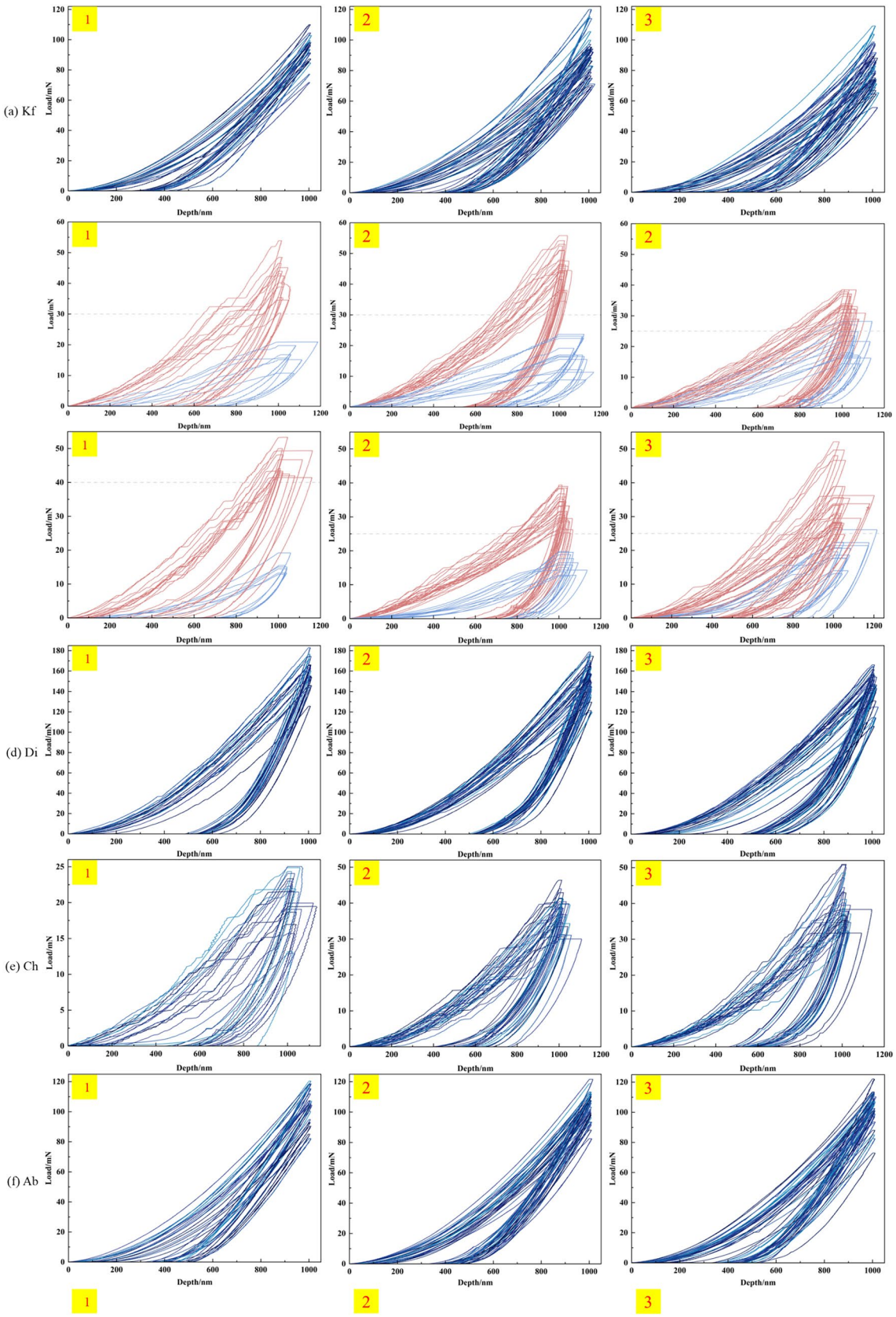


Fig. 7 Load–depth curves for different parts of different minerals under microwave irradiation; 1—untreated, 2—surface, 3—interior

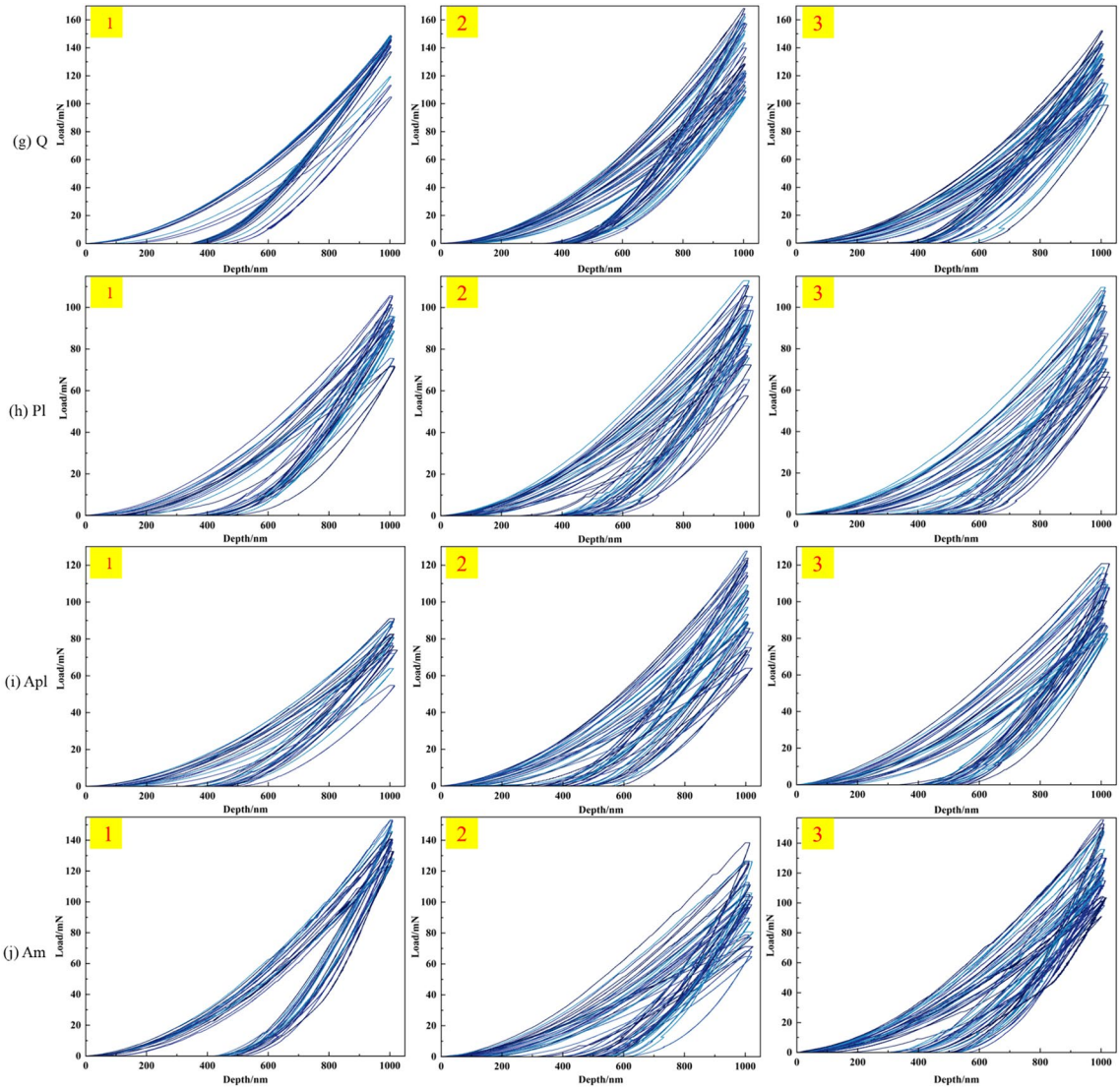


Fig. 7 (continued)

sample decreased significantly. The authors concluded that Am is more sensitive to microwaves, heats up faster, and melts internally. As a result, the interior of Am produces high-hardness melts in a dispersed manner (see Fig. 5j-3), leading to surround and base effects (Lu et al. 2020b; Wu et al. 2020; Yang et al. 2020a). This causes the maximum load and the residual indentation depth of the interior sample of Am to exhibit large values after microwave treatment.

Pop-in events during loading are characterized by sharp increases in indentation depth but small changes in load (Zhang et al. 2009). The main deformation mechanisms associated with pop-in events include crack formation, extension, or chip generation (Mikowski et al. 2008). As the number of indentations varied among samples, the frequency of pop-in events (f) was defined as the proportion of the total number of pop-in events (N_p) to the number of indentations (N), as shown in Eq. (7), to quantify the frequency of pop-in events for minerals:

$$f = \frac{N_p}{N} \tag{7}$$

The calculation results are displayed in Fig. 8, where quartz and feldspar minerals have the fewest number of pop-in events. The frequency of pop-in events for Q, Pl, and Apl is almost 0. The frequency of pop-in events for untreated samples and treated interior samples for Kf and Ab is about 0.2, and the frequency of pop-in events for treated surface samples is close to 0. The other two hard minerals, Di and Am, also had a relatively low frequency of pop-in events, 1.11 and 0.89 respectively, before microwave irradiation. After microwave treatment, the frequency of pop-in events

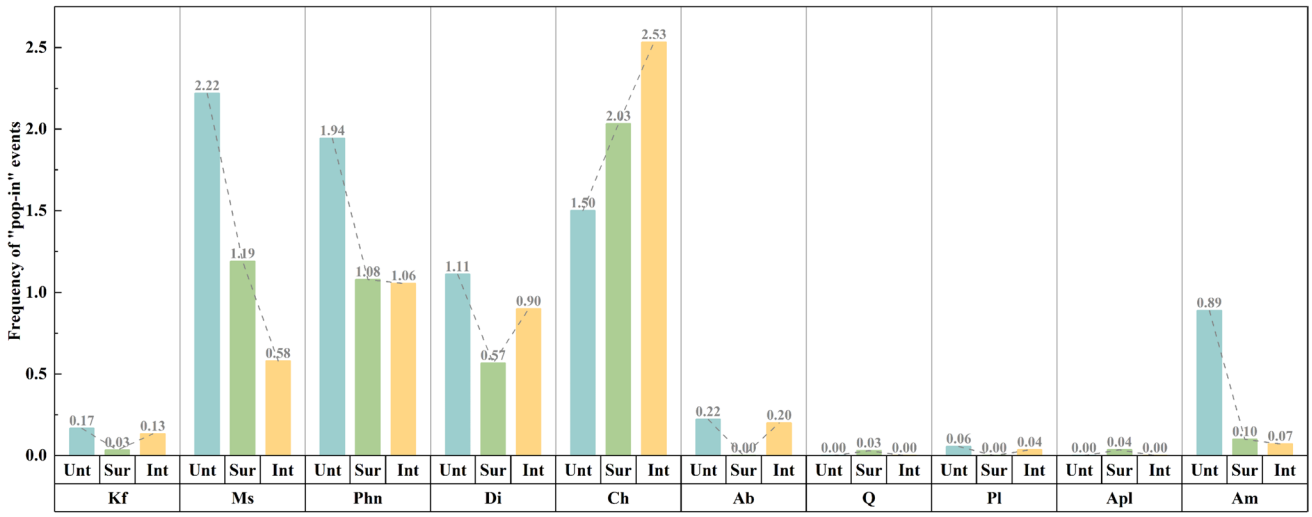


Fig. 8 Frequency of "pop-in" events for different parts of different minerals under microwave irradiation

decreased for surface and interior samples of both, with a more significant decrease observed in the Am samples. Ms and Phn exhibited the highest frequency of pop-in events at around 2.0, but microwave irradiation caused a significant decrease in the frequency of pop-in events for both mica minerals. The decrease in the number of pop-in events may be due to the increase in temperature after microwave action, which caused the samples to undergo a deformation transition from brittle to ductile (Espinoza et al. 2022). It is noteworthy that the pop-in events occurring in Ms, Phn, and Ch are more pronounced, i.e., the pop-in displacements are larger. Zhang et al. (2013) attribute them to kink-band generation and layer delamination. On the other side, the insignificant pop-in phenomenon produced by Q, Am, and the four feldspar minerals during loading was attributed to sudden dislocation nucleation and movement due to plastic deformation by Pathak and Kalidindi (2015) and Shim et al. (2008). Although Ch also has a pop-in event frequency as high as 1.50, it shows an abnormal increase after microwave treatment. Together with the decrease in the average residual indentation depth and the increase in the average value of the maximum load after microwave treatment, it can be concluded that Ch has been comprehensively hardened due to the microwave effect. On the other hand, while the mean value of the maximum load also showed an increase in Apl after microwave treatment, its average residual depth and the frequency of the pop-in event did not show the same anomaly as that of Ch. Therefore, the hardening mechanism of Apl may differ from that of Ch, which will be further analyzed later.

The holding stage, while eliminating creep effects (Mo et al. 2022), exhibits different creep properties in various rock-forming minerals. Therefore, the most direct parameter to quantify nanoindentation creep, maximum creep displacement (Ma et al. 2022), is also used in this work to describe the creep behavior of different minerals. Combining Table 2, Figs. 7 and 9, it can be observed that the creep displacement of Kf and Q is the smallest, at only around 5 nm. The creep displacement of Di, Ab, Pl, Apl, and Am is slightly larger, at approximately 10 nm. In addition, the three-layered silicate minerals exhibit the largest creep displacement, measuring about 50 nm. The results of Liu et al. (2022) for the creep behavior patterns of quartz, muscovite, and chlorite are similar. However, this paper disagrees with the conclusion of Mukherjee et al. (2023) that the creep displacement is proportional to the maximum load. One of the counterexamples is that the maximum load for Kf is much smaller than Di, but the creep displacement for Kf is also smaller than Di. After microwave heating, different minerals exhibited different trends in creep properties. The maximum creep deformation became more dispersed and generally increased, which Espinoza et al. (2022) attributed to the softening of the mineral structure in the rock due to the increase in temperature. The increment in the creep displacement of Di, Ab, Q, and Apl is relatively small, only about 1–3 nm. Curiously, the creep displacement of Apl's surface sample decreases after microwave heating. The increment in the creep displacement of Kf, Pl, and Am is larger, about 5–7 nm. Unusually, the increment in the creep displacement of Am's interior sample is much smaller than that of the surface sample after microwave treatment. Apparently, this is in the same sense as its anomalous residual indentation depths and maximum loads. Ms and Phn showed the largest increase in the creep displacement after microwave treatment, reaching over 10 nm. Attention should be drawn to the fact that a decrease in the value of creep displacement occurred in Phn's surface sample after microwave irradiation. In the opinion of the authors, this is mainly due to the

Table 2 Micro-mechanical parameters and heating properties of different parts of 10 rock-forming minerals under microwave action

Mineral	Sample treatment	Hardness/GPa	Young's modulus/GPa	Fracture toughness/MPa·m$^{1/2}$	Creep displacement/nm	ERP	EER	Poisson's ratio	H/E	H/K_c	Surface average temperature/°C	Average heating rate/°C/s	Dielectric loss factor	Hardness change rate	Young's modulus change rate
Kf	Unt	10.13 ± 1.31	65.49 ± 4.97	4.17 ± 0.53	5.78 ± 2.10	0.69 ± 0.05	0.75 ± 0.06	0.3	0.16 ± 0.02	2.47 ± 0.45	171.2	0.95	0.118	0.194	0.014
	Sur	8.16 ± 1.55	66.40 ± 11.05	3.86 ± 0.69	10.49 ± 4.46	0.59 ± 0.05	0.65 ± 0.06		0.12 ± 0.02	2.16 ± 0.50					
	Int	6.75 ± 1.58	61.50 ± 8.61	3.76 ± 0.88	12.43 ± 4.13	0.58 ± 0.07	0.62 ± 0.07		0.11 ± 0.02	1.87 ± 0.58					
Ms	Unt	1.92 ± 0.89	33.98 ± 12.75	1.96 ± 0.83	48.14 ± 41.83	0.47 ± 0.08	0.43 ± 0.08	0.25	0.06 ± 0.01	0.96 ± 0.14	318.7	1.77	0.136	0.026	0.423
	Sur	1.87 ± 0.84	48.36 ± 19.40	2.35 ± 1.02	56.97 ± 42.25	0.40 ± 0.06	0.35 ± 0.07		0.04 ± 0.01	0.79 ± 0.12					
	Int	1.42 ± 0.43	43.26 ± 10.49	1.89 ± 0.51	59.19 ± 32.18	0.35 ± 0.07	0.28 ± 0.04		0.10 ± 0.02	0.75 ± 0.11					
Phn	Unt	2.11 ± 1.09	33.58 ± 8.78	2.18 ± 1.01	52.50 ± 45.01	0.51 ± 0.16	0.46 ± 0.12	0.25	0.06 ± 0.02	0.95 ± 0.13	259.2	1.44	0.136	0.365	0.354
	Sur	1.34 ± 0.49	45.47 ± 15.52	1.87 ± 0.80	45.90 ± 23.77	0.31 ± 0.07	0.25 ± 0.04		0.03 ± 0.00	0.83 ± 0.56					
	Int	1.66 ± 0.77	27.66 ± 5.84	1.56 ± 0.61	69.77 ± 52.82	0.44 ± 0.15	0.41 ± 0.13		0.06 ± 0.02	1.05 ± 0.14					
Di	Unt	11.42 ± 1.32	174.73 ± 13.82	8.96 ± 1.02	8.95 ± 1.87	0.46 ± 0.03	0.47 ± 0.03	0.26	0.07 ± 0.00	1.27 ± 0.06	298.4	1.66	0.090	0.068	0.049
	Sur	10.64 ± 1.17	166.05 ± 21.62	8.71 ± 1.15	9.98 ± 2.44	0.47 ± 0.03	0.47 ± 0.03		0.06 ± 0.01	1.23 ± 0.09					
	Int	10.08 ± 1.50	177.29 ± 21.06	8.83 ± 1.43	11.22 ± 4.31	0.48 ± 0.05	0.46 ± 0.05		0.06 ± 0.01	1.15 ± 0.14					
Ch	Unt	1.10 ± 0.26	22.75 ± 3.56	1.28 ± 0.26	46.44 ± 31.05	0.44 ± 0.07	0.39 ± 0.07	0.27	0.05 ± 0.01	0.86 ± 0.11	277.2 (melt at 110 s)	2.52	-	1.027	0.568
	Sur	2.23 ± 0.39	38.50 ± 2.62	2.16 ± 0.43	27.31 ± 19.41	0.46 ± 0.09	0.44 ± 0.06		0.06 ± 0.01	1.05 ± 0.11					
	Int	2.21 ± 0.67	49.17 ± 7.09	2.29 ± 0.63	27.88 ± 26.16	0.45 ± 0.06	0.43 ± 0.10		0.05 ± 0.02	1.04 ± 0.43					
Ab	Unt	9.87 ± 1.69	76.49 ± 7.88	4.30 ± 1.24	7.50 ± 1.94	0.59 ± 0.06	0.67 ± 0.05	0.28	0.13 ± 0.02	2.42 ± 0.62	250.5	1.39	0.090	0.085	0.004
	Sur	9.03 ± 1.05	76.16 ± 6.24	4.58 ± 0.84	8.00 ± 1.99	0.59 ± 0.05	0.64 ± 0.04		0.12 ± 0.01	2.05 ± 0.49					
	Int	8.69 ± 1.29	75.92 ± 7.38	4.69 ± 0.64	8.51 ± 2.17	0.59 ± 0.05	0.63 ± 0.05		0.11 ± 0.01	1.88 ± 0.36					
Q	Unt	14.37 ± 1.98	102.24 ± 8.43	5.02 ± 0.41	4.40 ± 1.10	0.63 ± 0.04	0.73 ± 0.04	0.08	0.14 ± 0.01	2.86 ± 0.26	132.5	0.74	0.014	0.084	0.096
	Sur	13.16 ± 1.59	112.06 ± 6.85	6.06 ± 0.71	5.79 ± 1.69	0.61 ± 0.03	0.68 ± 0.03		0.12 ± 0.01	2.20 ± 0.37					
	Int	11.97 ± 2.40	96.15 ± 7.38	4.70 ± 0.93	7.21 ± 5.74	0.60 ± 0.07	0.69 ± 0.08		0.12 ± 0.02	2.63 ± 0.68					
Pl	Unt	7.69 ± 1.25	67.36 ± 6.24	4.68 ± 0.53	10.97 ± 3.45	0.61 ± 0.05	0.63 ± 0.05	0.29	0.11 ± 0.01	1.65 ± 0.23	188.2	1.05	0.039	0.056	0.021
	Sur	7.26 ± 1.66	65.89 ± 8.46	4.61 ± 0.50	15.33 ± 6.36	0.60 ± 0.07	0.60 ± 0.09		0.11 ± 0.02	1.58 ± 0.34					
	Int	6.40 ± 1.30	63.02 ± 7.31	4.56 ± 0.83	14.51 ± 5.61	0.61 ± 0.09	0.59 ± 0.07		0.10 ± 0.02	1.42 ± 0.24					
Apl	Unt	6.88 ± 1.24	56.62 ± 5.34	3.50 ± 0.56	11.56 ± 4.31	0.60 ± 0.05	0.66 ± 0.06	0.29	0.12 ± 0.02	2.00 ± 0.43	145.5 (burst at 45 s)	3.24	-	0.238	0.202
	Sur	8.52 ± 2.68	68.06 ± 11.65	4.00 ± 0.85	10.15 ± 5.40	0.60 ± 0.07	0.66 ± 0.08		0.12 ± 0.02	2.17 ± 0.65					
	Int	7.75 ± 1.54	70.68 ± 10.77	4.19 ± 0.53	14.72 ± 7.08	0.57 ± 0.07	0.62 ± 0.07		0.11 ± 0.02	1.89 ± 0.49					
Am	Unt	11.56 ± 1.00	117.88 ± 15.26	6.74 ± 0.90	7.85 ± 2.26	0.54 ± 0.03	0.59 ± 0.04	0.29	0.10 ± 0.02	1.75 ± 0.32	250.1 (melt at 82 s)	3.05	0.324	0.292	0.327
	Sur	8.19 ± 1.77	79.30 ± 22.62	4.23 ± 1.30	15.41 ± 6.08	0.53 ± 0.08	0.59 ± 0.11		0.11 ± 0.03	2.11 ± 0.75					
	Int	11.42 ± 1.89	104.04 ± 26.47	6.00 ± 1.43	8.73 ± 3.45	0.58 ± 0.09	0.63 ± 0.10		0.12 ± 0.03	2.04 ± 0.75					

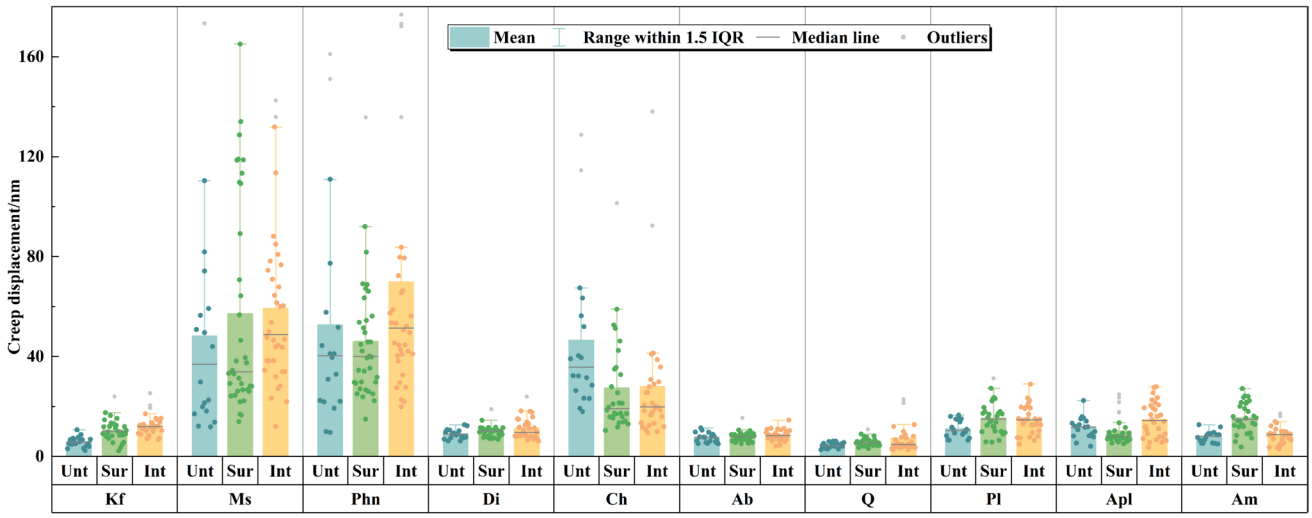


Fig. 9 Creep displacement of different parts of different minerals under microwave irradiation

reduction in the number of apparent mutations in their holding phase, which is thought to result in a slowdown in crack extension and a reduction in chips due to the decreased brittleness of the minerals after microwave treatment. Besides, Ch once again showed the opposite experimental phenomenon, with a significant reduction in creep displacement of about 20 nm in both surface and interior samples after microwave irradiation. This finding further confirms the hardening of chlorite under microwave irradiation.

Remarkably, the standard deviations of all mechanical indices of Ms and Phn are on the high side. Meanwhile, as shown in Fig. 7b, c, the load–depth curves of the two mica minerals present two parts of a split, which is the variability caused by the different angles between the indentation direction and the disintegration direction of the mica minerals. Observing Table 4, it can be seen that the hardness and Young's modulus values of mica minerals measured by different scholars are also quite diverse. Lanin et al. (2021) conducted a specific study for this purpose and measured the Young's modulus of biotite as 35 GPa for the direction perpendicular to the lamination, while the Young's modulus for the direction parallel to the lamination is 95 GPa. In the present paper, the mechanical properties of mica minerals are tested for different indentation directions, and the numerical patterns of the results are completely opposite to theirs. The reason for this is that the spherical indenter used by Lanin et al. and the Berkovich indenter used by the authors are different. However, the differences in the mechanical properties due to the different angles between the indentation direction and the disintegration direction should be recognized. As shown in Fig. 7b, c, when indenting parallel to the solvation direction, the maximum load of the mica minerals is smaller, the residual indentation depth is larger, the creep displacement in the holding phase is larger, the frequency of the pop-in event is smaller, and the loading segment of the curves is smoother. The performance is completely opposite when indenting perpendicular to the solvation direction. After microwave irradiation, the maximum load of the curve perpendicular to the solvation direction significantly decreases. In addition, the residual depth increases, the creep displacement increases, and the frequency of pop-in events decreases. On the other hand, the creep displacement, residual depth, and frequency of pop-in events of the indentation curves parallel to the solvation direction showed consistent trends with those of the curves perpendicular to the solvation direction. Nevertheless, the maximum load varied differently in different parts. Specifically, in Table 3, the surface samples of Ms and Phn showed a decrease in hardness in both parallel and perpendicular directions to the solvation direction, with a higher degree of decrease in the perpendicular direction to the solvation direction. However, the interior samples showed a modest increase in hardness in the direction parallel to the disintegration and a decrease in hardness in the direction perpendicular to the disintegration. Therefore, after microwave irradiation, the surface samples of mica minerals exhibit a reduced degree of severance between the two parts of the load–depth curves, while the interior samples of the minerals show a better fusion phenomenon in the curves of the two parts.

3.3 Microwave Effects on the Hardness and Young's Modulus

Liu et al. (2022) stated that the elastic modulus represents the resistance to elastic deformation, while hardness represents the resistance to plastic deformation in indentation experiments. Since the indentation depth was set to be

Table 3 Micro-mechanical parameters of different parts of Ms and Phn under microwave irradiation with different indentation directions

Mineral	Indentation direction	Sample treatment	Hardness/GPa	Young's modulus/GPa	Fracture toughness/MPa·m$^{1/2}$	Creep displacement/nm	ERP	EER	H/E	H/K_c
Ms	Vertical cleavage	Unt	2.47 ± 0.44	41.91 ± 6.02	2.45 ± 0.50	26.39 ± 15.31	0.49 ± 0.07	0.46 ± 0.07	0.06 ± 0.01	1.02 ± 0.12
		Sur	2.42 ± 0.31	61.05 ± 7.49	3.02 ± 0.47	29.92 ± 10.56	0.42 ± 0.03	0.37 ± 0.04	0.04 ± 0.00	0.81 ± 0.09
		Int	1.66 ± 0.29	49.37 ± 6.56	2.14 ± 0.38	45.73 ± 22.24	0.35 ± 0.06	0.30 ± 0.04	0.03 ± 0.01	0.78 ± 0.09
	Parallel cleavage	Unt	0.81 ± 0.19	18.14 ± 4.61	0.98 ± 0.23	91.63 ± 45.04	0.42 ± 0.09	0.37 ± 0.07	0.05 ± 0.01	0.83 ± 0.07
		Sur	0.75 ± 0.24	22.98 ± 7.76	1.03 ± 0.30	111.07 ± 28.62	0.35 ± 0.06	0.29 ± 0.08	0.04 ± 0.01	0.74 ± 0.16
		Int	0.94 ± 0.22	31.03 ± 3.91	1.38 ± 0.33	86.09 ± 32.82	0.35 ± 0.09	0.26 ± 0.04	0.03 ± 0.01	0.70 ± 0.12
Phn	Vertical cleavage	Unt	2.80 ± 0.53	39.22 ± 3.46	2.84 ± 0.37	58.47 ± 54.53	0.60 ± 0.11	0.52 ± 0.09	0.07 ± 0.01	0.98 ± 0.10
		Sur	1.64 ± 0.26	54.91 ± 6.84	2.36 ± 0.38	36.54 ± 13.32	0.34 ± 0.05	0.27 ± 0.04	0.03 ± 0.00	0.70 ± 0.08
		Int	2.03 ± 0.68	29.31 ± 4.90	1.89 ± 0.48	59.49 ± 45.97	0.52 ± 0.10	0.49 ± 0.08	0.07 ± 0.01	1.07 ± 0.16
	Parallel cleavage	Unt	0.74 ± 0.11	22.31 ± 2.59	0.87 ± 0.26	40.56 ± 9.45	0.34 ± 0.08	0.33 ± 0.05	0.03 ± 0.00	0.89 ± 0.18
		Sur	0.73 ± 0.12	26.59 ± 9.26	0.89 ± 0.39	64.61 ± 29.21	0.26 ± 0.07	0.23 ± 0.04	0.03 ± 0.01	1.10 ± 0.94
		Int	0.92 ± 0.13	24.38 ± 6.39	0.91 ± 0.12	90.33 ± 61.37	0.27 ± 0.06	0.26 ± 0.04	0.04 ± 0.01	1.02 ± 0.09

identical, meaning the indentation area was the same, the only parameter influencing the hardness was the maximum load. Thus, the experimental patterns and variation patterns under microwave effects of mineral hardness are extremely similar to those of the maximum loads in Sect. 3.1. From Table 2 and Fig. 10, the hardnesses (maximum load, hardness, Young's modulus, etc., all representing their average values) of Di, Q, and Am are larger, with values of 11.42 GPa, 14.37 GPa, and 11.56 GPa, respectively. Following with slightly lower hardnesses are the four feldspar minerals, Kf, Ab, Pl, and Apl, with hardness values of, in descending order, 10.13 GPa, 9.87 GPa, 7.69 GPa, and 6.88 GPa. As with Liu et al. (2022), Ms, Phn, and Ch were classified as soft minerals due to the presence of water molecules in the interlayers of layered silicate minerals (Mukherjee and Misra 2023), with hardnesses of 1.92 GPa, 2.11 GPa, and 1.10 GPa, respectively. Obviously, the classification and ordering of the minerals obtained according to the maximum load and hardness are identical. To verify the reasonability of the test results, the authors have compiled the values of mineral hardness, Young's modulus, and fracture toughness tested in existing studies in Table 4. Comparisons revealed that the hardness values tested in that study were almost always within the range of, or similar to, the results of other scholars' tests. As illustrated in Table 3, the hardness of Ms and Phn perpendicular to the solvation direction was measured to be 2.47 GPa and 2.80 GPa, respectively, taking into account the angle between the indentation direction and the solvation direction. On the other hand, the hardness obtained in the direction parallel to the disintegration direction is 0.81 GPa and 0.74 GPa, respectively. Apparently, it is similar to the test results of the maximum loads of both: the maximum loads of Ms and Phn are smaller when the indenter is pressed parallel to the lamination direction. It can be seen that Ms and Phn are harder in the direction perpendicular to the disintegration direction and more resistant to plastic deformation. The electrostatic interactions between layers and interlayers of cations, cation hydration, and Born repulsive forces are the main physical phenomena that directly affect the mechanical behavior of such minerals, resulting in the mica anisotropy in the interlayer parallel and interlayer normal directions of the minerals. It can also be noticed from Table 4 that the hardness of mica minerals from Zhu et al. (2007) and Liu et al. (2022) is close to the hardness in the perpendicular solvation direction, whereas the results from Komadja et al. (2022) and Lei et al. (2021b) are similar to the results when the indentation direction is not taken into account.

After microwave irradiation, the trend of hardness was similar to the maximum load. As shown in Table 2, the weakening effects of Di, Ab, Q, and Pl were mediocre, with the hardness of the surface samples decreasing by 5.61–8.62%, and the interior samples decreasing even

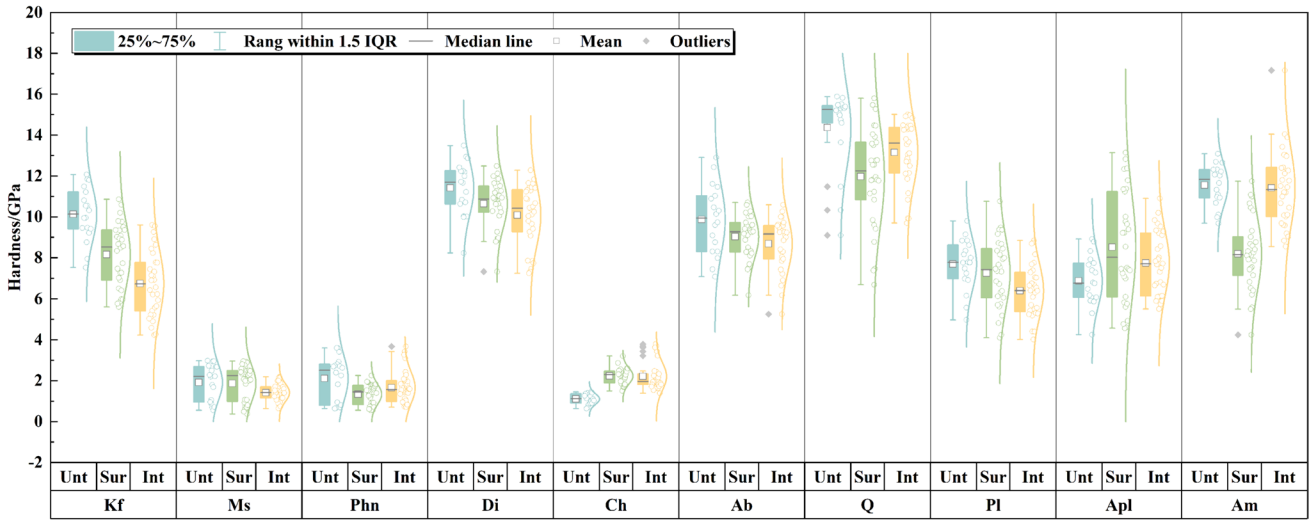


Fig. 10 Hardness of different parts of different minerals under microwave irradiation

Table 4 Nanoindentation hardness, Young's modulus and fracture toughness of the primary rock-forming minerals in granite at room temperature

Mineral	Hardness/GPa	Hardness (this work)/GPa	Young's modulus/GPa	Young's modulus (this work)/GPa	Fracture toughness/MPa·m$^{1/2}$	Fracture toughness (this work)/MPa·m$^{1/2}$	References
K-feldspar(Kf)	8.3 11.4 ± 0.4	10.1 ± 1.3	71.4 100.6 ± 1.2	65.5 ± 5.0	– 4.8 ± 0.2	4.7 ± 1.2	Maruvanchery and Kim (2020) Liu et al. (2023)
Muscovite(Ms)	3.3 ± 0.6 1.9 ± 1.0 1.8 ± 0.8 2.7 –	1.9 ± 0.9 2.1 ± 1.1[a]	59.2 ± 8.1 21.4 ± 8.1 25.9 ± 8.3 55.6 –	34.0 ± 12.8 33.6 ± 8.8[a]	– – – – 2.12	2.5 ± 1.1 2.7 ± 1.4[a]	Zhu et al. (2007) Komadja et al. (2022) Lei et al. (2021b) Liu et al. (2022) Lei et al. (2021a)
Diopside(Di)	5.9 ± 0.3[b] 12.0	11.4 ± 1.3	– 210.1	174.6 ± 13.8	– 7.0	10.8 ± 2.2	Dorner and stockhert (2004) Meng et al. (2022)
Chlorite(Ch)	1.5	1.1 ± 0.26	37	22.7 ± 3.6	–	1.5 ± 0.5	Liu et al. (2022)
Albite(Ab)	– 5.5 ± 0.4 7.5 ± 0.1	9.9 ± 1.7	69.0 69.9 ± 0.9 71.8 ± 0.4	76.5 ± 7.9	– – –	4.4 ± 2.1	Mavko et al. (2020) Ayatollahi et al. (2020) Sousa et al. (2014)
Quartz(Q)	15.1 12.45 ± 3.5 13.0 ± 0.7 12.6 14.5 ± 0.9	14.4 ± 2.0	96.8 128.9 ± 41.7 104.2 ± 5.9 110.9 106.3 ± 2.9	102.2 ± 8.4	– – – – 5.3 ± 0.1	6.7 ± 1.8	Tang et al. (2022) Komadja et al. (2022) Zhu et al. (2007) Mo et al. (2022) Liu et al. (2023)
Plagioclase(Pl)	10.6 8.4 9.0 3.2 ± 0.5	7.7 ± 1.2 6.9 ± 1.2[c]	69.5 79.2 71.1 67.2 ± 6.4	67.4 ± 6.2 56.6 ± 5.3[c]	– – 3.3 2.7 ± 0.3	4.8 ± 1.2 3.7 ± 1.1[c]	Tang et al. (2022) Maruvanchery and Kim (2020) Meng et al. (2022) Liu et al. (2023)
Amphibole(Am)	– 12.3 12.0	11.6 ± 1.0	141.8–157.1 155.0 156.4	117.9 ± 15.3	– – 6.5	8.1 ± 1.0	Brown and Abramson (2016) Mo et al. (2022) Meng et al. (2022)

a. Phengite; b. Hardness at 300℃; c. Altered plagioclase

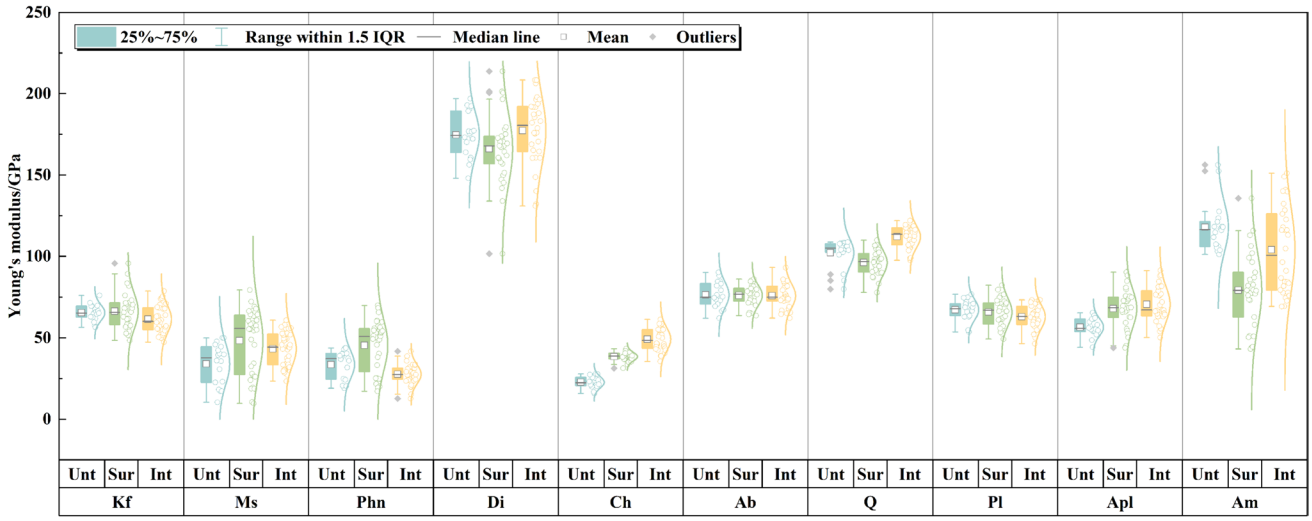


Fig. 11 Young's modulus of different parts of different minerals under microwave irradiation

more, to 11.7–16.8%. The weakening of Kf and Am is better, with discount rates of up to 30% or more. Amphibole also weakened better at high temperatures than quartz and feldspar, according to the results of Mo et al. (2022). In line with the anomalies of residual depth and creep displacement, the discount rate of hardness for Am's interior sample is even lower, at only 1.21%, which contradicts common perception. This is due to amphibole, one of the microwave-sensitive minerals (Kahraman et al. 2022; Lu et al. 2017), being heated as its internal melting temperature is much higher than the dehydration temperature of 800 °C (Riecker and Rooney 1969). This further justifies the explanation given in Sect. 3.1 regarding the surround and base effects of melts in the interior sample of Am. For Ms and Phn, as shown in Table 2, the discounting patterns of the hardness of both after microwave action are very similar to those of the maximum load in Sect. 3.1. That is, the reduction of Ms is very minor, while the reduction of Phn is large. However, when the angle between the indentation direction and the lamination direction is taken into account, the results obtained are shown in Table 3, which are quite different from those shown in Table 2. Specifically, there is a minor decrease in the hardness of the surface samples of both in the direction parallel to the disintegration, while the hardness of the interior samples increases instead. In other words, there is a hardening of the interior samples of both in the direction parallel to the laminations. This may be due to the loss of binding water and tectonic water at high temperatures, resulting in the contraction of the spacing along the disintegration (Zhang et al. 2018). Similar phenomena of heat-treatment strengthening of mica minerals were found by Fan et al. (2018) and Mo et al. (2022). Moreover, the indentation morphology diagrams of Mo et al. further verified that their indentation direction was exactly parallel to the direction of disintegration. Observation of a thin section of Apl (see Fig. 6) revealed that the feldspar minerals underwent localized carbonization, resulting in the formation of trace amounts of carbonate. Therefore, it is believed that the increased hardness in Apl is due to the decomposition of carbonates at high temperatures, generating thermal expansion air pressure. This pressure then leads to the closure of microcracks within the mineral samples, ultimately increasing the hardness of the minerals. Liu (2018) also found that the carbonate-filled amygdaloid would decompose at high temperatures, generating air pressure that would blow up the rock samples.

As presented in Table 2, Figs. 10 and 11, the distributions of Young's modulus and hardness of the 10 rock-forming minerals basically conform to normal distributions, and their means and standard deviations are denoted as error bars. Among them, the mechanical parameters of mica minerals are more discrete, which is due to their special solvation structure and the influence of other impurities. Comparison of Tables 2 and 3 shows that the standard deviation of both Ms and Phn decreases, and the dispersion of hardness and Young's modulus decreases significantly when considering the angle between the indentation direction and the disintegration. After the microwave effect, the dispersion of mechanical parameters of Kf, Di, Ch, Pl, Apl, and Am, a total of six minerals, further increased. This can be attributed to the fact that microwave effects cause an increase in microdefects in minerals, such as cracks and pores. The dispersion of the distributions of hardness and Young's modulus did not change significantly for Ab and Q, and there was even a slight decrease in the anomalous phenomenon. As analyzed by the authors, this can be attributed partly to the better mechanical properties of both materials and partly to their higher purity and relatively better homogeneity,

making them less susceptible to impurities. For Ms and Phn, the standard deviation of Young's modulus undergoes a significant increase after microwave irradiation, even in the same indentation direction, especially for their surface samples. The same phenomenon is also found in the surface samples of Kf, Di, Pl, and Apl. It is speculated that this may be related to the poorer homogeneity of the mineral surface temperature distribution due to the inhomogeneous distribution of the electric field. Intriguingly, the dispersion of hardness parameters for Ms and Phn exhibits a decrease perpendicular to the direction of disintegration and an increase parallel to the direction of the laminations. The authors attribute this, on the one hand, to the fact that mica is a transversely isotropic sheet silicate, possessing different interlayer cations and interlayer thicknesses from layer to layer (Mukherjee and Misra 2023), leading to greater inhomogeneity of mica minerals parallel to the solvation direction. On the other hand, under microwave irradiation, thermal cracks in mica minerals are more likely to be formed in the interlayers, as suggested by Mo et al. (2022).

Both Liu et al. (2023) and Ante et al. (2018) found a statistically significant linear correlation between the hardness of minerals and Young's modulus on the nanoscale. Their findings are also consistent with the general rule that materials with higher Young's modulus have higher contact hardness. A similar finding is observed in this article, as demonstrated in Table 2 and Fig. 11, where the minerals Di, Q, and Am, which are harder, also possess higher Young's moduli of 174.73 GPa, 102.24 GPa, and 117.88 GPa, respectively. As with the hardness ordering, the four feldspar minerals have slightly lower Young's modulus, ranging from 55 to 80 GPa. The three layered silicates Ms, Phn, and Ch possess the lowest Young's moduli of 33.98 GPa, 33.58 GPa, and 22.75 GPa, respectively. The Young's modulus of the minerals tested in this study is similar to the results of other scholars in Table 4, confirming the validity of the test results. For Ms and Phn, combined with Table 3, it was found that, similar to the hardness, Young's modulus perpendicular to the disintegration direction was significantly greater than that parallel to the disintegration direction. Almost all of the Young's modulus of Kf, Ab, Q and Pl showed a slight decrease after microwave treatment. Surprisingly, the Young's modulus of the surface samples of Kf and Q showed an increase. This can be attributed to the lower surface temperatures of both samples, resulting in an expansion of the mineral's lattice spacing due to the temperature rise. A similar observation of an augmented Young's modulus in quartz and feldspar at low temperatures was also reported by Mo et al. (2022). However, as the temperature reaches 200 °C–300 °C, the displacement phase transition alters the crystallinity of plagioclase and decreases Young's modulus (Mo et al. 2022). This view is verified by the experiment's observation of a decrease in Young's modulus of feldspar samples at higher temperatures. The Young's modulus of Di also decreases slightly, but the Young's modulus of its interior sample instead increases by 1.47%. This relatively minor difference is attributed by the authors to the interference of impurities and inclusions, or to the influence of the surface roughness. The Young's modulus of Am decreases by a larger amount, but the interior sample likewise decreases by a much smaller amount, which is similar to the pattern of change in hardness and residual depth of Am under microwave action. It is also attributed to the surround and base effects of the internal melt. Consistent with the variation in hardness, Young's modulus of the enhanced minerals Ch and Apl also increases. Curiously, Young's modulus of Ms and Phn also display an increase in both indentation directions.

3.4 Microwave Effects on the Fracture Toughness

The level of fracture toughness reflects the energy required for cracking a rock or mineral, and minerals with lower fracture toughness are more likely to produce complex crack networks (Ma et al. 2020). Therefore, accurately measuring the fracture toughness of minerals under microwave irradiation is important for predicting the extent of crack expansion when granite is heated. The energy analysis method proposed by Cheng et al. (2002) establishes a new relationship between energies. The method is not affected by the hardness and Young's modulus of materials and is applicable to both depressed and raised cases (Lei 2022). In addition, the fracture toughness can be calculated using only the single load–depth curve of the nanoindentation. As depicted in Fig. 2, during nanoindentation, the total energy (U) is composed of two portions: elastic energy (U_e) and plastic energy (U_p). The plastic energy (U_p), in turn, consists of two portions: induced fracture energy (U_{frac}) and pure plastic energy (U_{pp}). The specific energy balance equation is expressed as follows (Cheng et al. 2002; Ma et al. 2020):

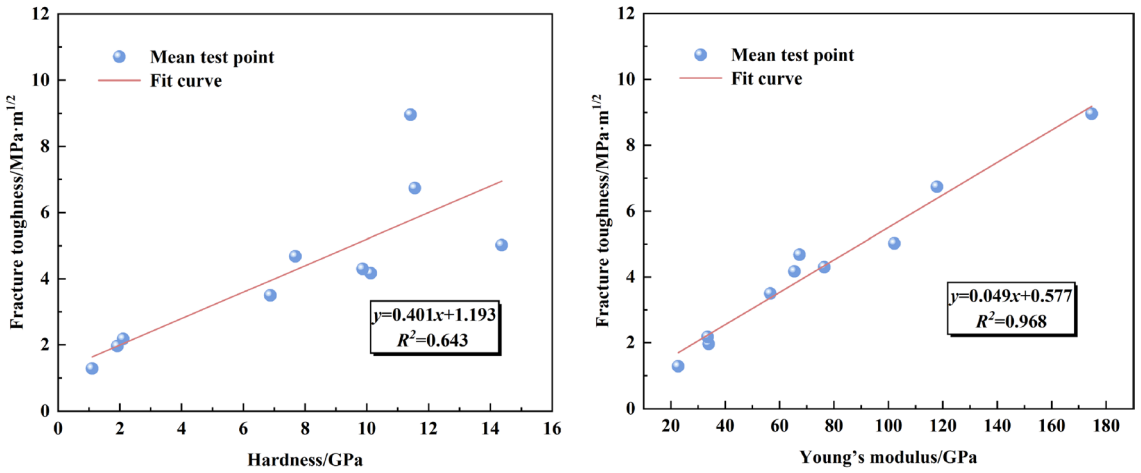
$$U = U_e + U_p = U_e + U_{pp} + U_{frac} \tag{8}$$

where U and U_e can be gained by integrating the loading and unloading curves over the load–displacement curve.

According to the linear elastic fracture mechanics, the fracture toughness K_{IC} can be calculated for plane strain conditions:

$$K_{IC} = \sqrt{G_c E_r \left(1 - \nu^2\right)} \tag{9}$$

This equation applies to thick boards, such as rocks and minerals, where the constraining effect of deformation in the thickness direction results in no strain in the z direction, i.e., $\varepsilon_z = 0$. This state, where strain exists in only two directions, is called the plane strain case. Here, E_r represents the



(a) Relationship between fracture toughness and hardness (b) Relationship between fracture toughness and Young's modulus

Fig. 12 Fitting of fracture toughness to hardness and Young's modulus of 10 minerals

reduced modulus and G_c represents the strain energy release rate. Gc can be calculated using the following equation:

$$G_c = \frac{\partial U_{frac}}{\partial A_c} = \frac{U_{frac}}{A_c} \tag{10}$$

where A_c is the projection contact area and U_{frac} can be calculated from Eq. (8), expressed as follows:

$$U_{frac} = U_p - U_{pp} = U - U_e - U_{pp} \tag{11}$$

Although U and U_e are known, U_{pp} cannot be directly determined from curves. It is typically obtained through a combination of experimental and finite element simulation results. According to Cheng et al. (2002), U_{pp} can be expressed as

$$U_{pp} = \left\{1 - \left[\frac{1 - 3\left(\frac{h_f}{h_{\max}}\right)^2 + 2\left(\frac{h_f}{h_{\max}}\right)^3}{1 - \left(\frac{h_f}{h_{\max}}\right)^2}\right]\right\} U \tag{12}$$

The fracture toughness can be obtained by substituting it into Eqs. (11), (10), and (9) sequentially. In this process, it is not necessary to measure the length of the crack.

Prior to this, Lei (2022) found a well-fitted linear relationship between hardness and fracture toughness. However, more scholars (Lawn and Wilshaw 1975; Lei 2022; Liu et al. 2016, 2023) have since derived a positive correlation between fracture toughness and Young's modulus. In other words, minerals with high Young's modulus tend to have relatively large fracture toughness values. This is because an increase in Young's modulus can enhance the extreme fracture strength and fracture resistance of the material (Yuan and Xi 2011).

To verify the validity of both views, as shown in Fig. 12, this study plotted the hardness and Young's modulus of minerals versus fracture toughness based on the hardness, Young's modulus, and fracture toughness of the 10 minerals tested in the experiment, respectively. It can be found that although the fracture toughness tends to increase with increasing hardness overall, the distribution of mean points gradually disperses with increasing hardness. In addition, the coefficient of determination, R^2, is only 0.654, which at least suggests that the relationship between hardness and fracture toughness is not purely linear. The reason why Lei (2022) came up with a better linear fit may be that they did not test a sufficiently large number of species of rock-forming minerals. Conversely, the linear fit of Young's modulus to fracture toughness is excellent, with a coefficient of determination, R^2, as high as 0.980. Thus, the authors confirmed the positive correlation between Young's modulus and fracture toughness by measuring the micromechanical parameters of a wider variety of rock-forming minerals. Specific results are shown in Table 2 and Fig. 13. Di, Q, and Am, which have the highest Young's modulus, also have the highest fracture toughness of 10.33, 6.46, and 8.03 MPa•m$^{1/2}$, respectively, followed by four feldspar minerals: Kf, Ab, Pl, and Apl, with fracture toughnesses in the range of 3.5–5.0 MPa•m$^{1/2}$. Layered silicate minerals, Ms, Phn, and Ch, possess the lowest fracture toughness values of 2.57, 2.71, and 1.55 MPa•m$^{1/2}$, respectively. By considering the angle between the indentation direction and disintegration direction, it is observed that in the perpendicular disintegration direction, the fracture toughness of Ms and Phn is greater, reaching 3.19 and 3.60 MPa•m$^{1/2}$, respectively. In the parallel disintegration direction, the fracture toughness of Ms and Phn is only 1.33 and 0.94 MPa•m$^{1/2}$, respectively. This indicates that Ms and

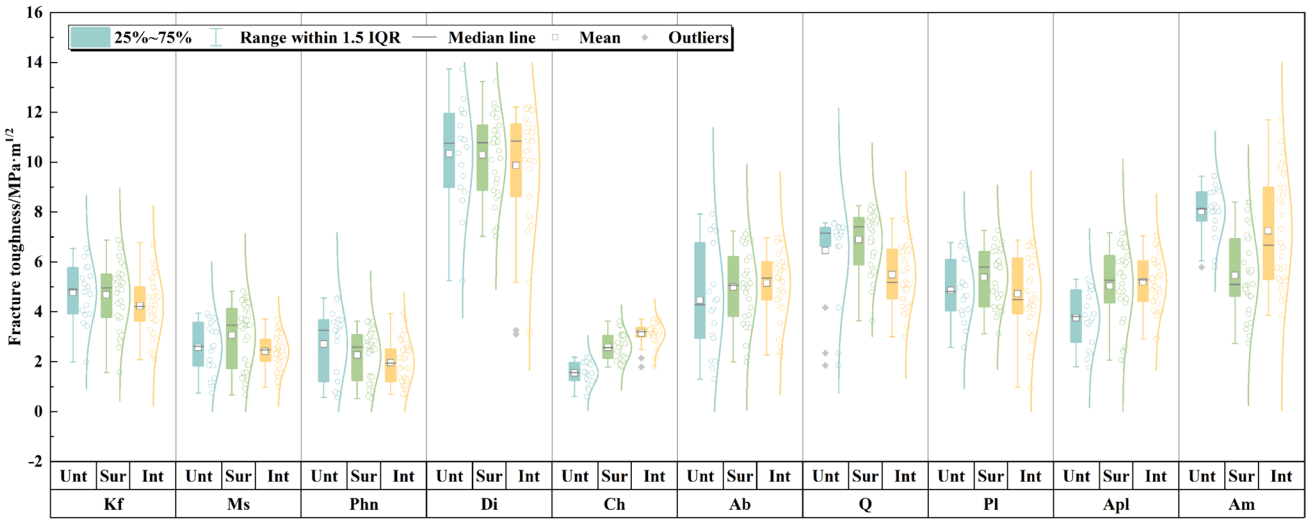


Fig. 13 Fracture toughness of different parts of different minerals under microwave action

Phn are more resistant to crack extension in the perpendicular disintegration direction.

Following exposure to microwave irradiation, the fracture toughness of both surface and interior samples of Ch and Apl increased, suggesting that the microwave effect strengthened the resistance to crack propagation in both materials. Notably, the fracture toughness of the interior sample of Ch experienced a significant enhancement of 78.9%. As shown in Table 3, similar to the variation pattern of the maximum load and hardness of mica, the fracture toughness decreases in the direction perpendicular to the disintegration after microwave irradiation, while the fracture toughness increases slightly in the direction parallel to the disintegration. The fracture toughness of Kf, Di, Q, Pl, and Am all decreased, with Di and Pl decreasing less, by within 3%. This is followed by Q and Kf with a decrease of more than 5%. The surface sample of Am showed the highest reduction in fracture toughness, which amounted to 37.24%. However, the surround and base effects of the melt also resulted in a smaller reduction in fracture toughness of 10.98% for the interior sample of Am. Interestingly, the fracture toughness of the surface sample of Q increased by 20.72%. This phenomenon was also found in the variation pattern of Young's modulus of the surface sample of Q, which was attributed to the intensification effect of quartz minerals at low temperatures. In addition, as a feldspar mineral, Ab also experienced a small increment in fracture toughness but a slight reduction in both hardness and Young's modulus. Therefore, if the small increase in fracture toughness of Ab is also categorized as a low-temperature strengthening effect in the same way as Q, a corollary to the occurrence of low-temperature strengthening in quartz and feldspar minerals is that there may be a dual tendency to changes in different micromechanical parameters of different minerals. In other words, there may be a combination of synergistic or competing mechanisms at low-temperature heat treatment. The authors propose that there are two main mechanisms involved in this low-temperature strengthening effect. First, the evaporation of water in different states enhances the effective stress of the rock (Sirdesai et al. 2017) and the friction between mineral grains (Kissell 1972; Zhao et al. 2017; Zuo et al. 2017), and also creates more free space for grain expansion (Sirdesai et al. 2017; Zhang et al. 2014). Second, the initial expansion of the grains closes the existing microcracks and makes the structure tight (Tian et al. 2014; Wong et al. 2020; Yang et al. 2017). Naturally, the reinforcing effects of thermal expansion are governed by a variety of factors. For instance, Malaga et al. (2006) found that the intense heterogeneity and complexity of the structure may render the particles more difficult to expand and less likely to be strengthened. Funatsu et al. (2004) and Yavuz et al. (2010) found that the compaction of minerals with porous structures was more pronounced and that new cracks sprang up at higher temperatures. In addition, Fei (1995) found that quartz had the largest coefficient of thermal expansion among rock-forming minerals, with a bulk expansion about four times that of feldspar (Siegesmund et al. 2008). Taken together, quartz has a shorter low-temperature strengthening phase due to its dense structure, low water content, and high coefficient of thermal expansion, while feldspar has a relatively longer low-temperature strengthening phase. This was demonstrated in the experiments conducted by Mo et al. (2022), where it was observed that the reduced modulus of quartz begins to decline at temperatures exceeding 100 °C, whereas the reduced modulus of feldspar starts to decline at temperatures exceeding 300 °C. Yet, due to the superposition of other known or unknown synergistic or competing mechanisms, the trends of other mechanical parameters are not identical, such as the hardness of quartz, which similarly

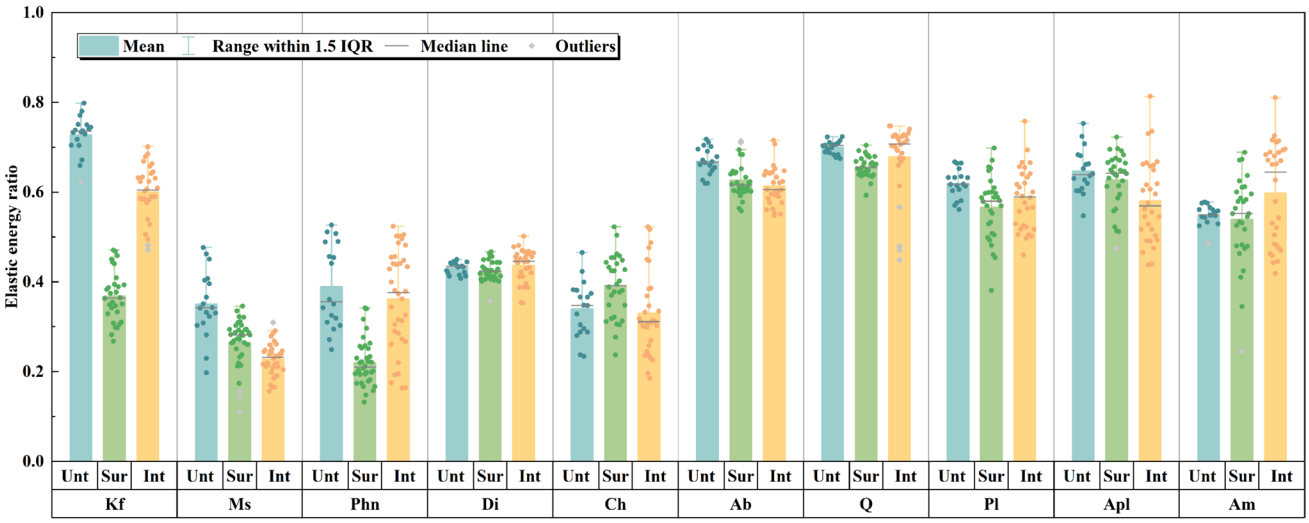


Fig. 14 Elastic energy ratio of different parts of different minerals under microwave action

starts to decrease at 100 °C, but the hardness of feldspar shows no sign of increasing at all in the study of Mo et al. (2022). Consequently, the interaction and dominance of the various mechanisms in different cases need to be further investigated.

3.5 Microwave Effects on Deformation–Destruction Characteristics

As Eq. (8) shows, the total energy consists of elastic energy, pure plastic energy and induced fracture energy. However, during unloading, only the elastic energy is released, so the elastic energy can be acquired by calculating the area enclosed by the unloading curve and the *x*-axis. Hence, this paper introduces the Elastic Recovery Parameter (ERP) as depicted in Eq. (13) and the Elastic Energy Ratio (EER) as illustrated in Eq. (14), which have been widely employed by numerous scholars (Ivanova et al. 2018; Liu et al. 2022; Shushkov et al. 2022), to elucidate the elastic recovery capability of minerals when subjected to microwave irradiation. The objective is to establish a connection between mineralogy and the response of elastic–plastic deformation.

$$ERP = \frac{h_{max} - h_f}{h_{max}} \tag{13}$$

$$EER = \frac{U_e}{U_t} \tag{14}$$

Lu et al. (2022) demonstrated through experiments that higher strength minerals lead to better recovery of indentation work, and they also found that high-strength minerals are more resistant to damage. Similar conclusions are found in this work, as shown in Fig. 14 and Table 2, where the higher strength Q, Am with feldspar minerals Kf, Ab, Pl and Apl almost all have EER averages of 0.59 or more. The low strength coupled with the orientation-preferred stacking arrangement leads to mean EER values below 0.46 for the layered silicate minerals Ms, Phn and Ch, i.e. plastic behavior plays a dominant role in the mechanical response of the three minerals. As espoused by Liu et al. (2022), this differential preferential elastic or plastic character is related to minerals with differing crystal structures. Quartz and feldspar minerals are silicate minerals with shelf-like crystal structures, while muscovite and chlorite are silicate minerals with lamellar structures. Interestingly, Di, despite having greater hardness and the highest Young's modulus, has a mean EER value of only 0.47, which is also related to the crystal structure of the minerals. Although both amphibole and diopside are silicates with chain crystal structures, diopside has a single-chain structure with short columnar single crystals and dense massive aggregates. On the other hand, amphibole has a double-chain structure with long columnar single crystals and columnar or radial aggregates, which have better always lengthening properties. This might be the fundamental reason for the difference in the elastic–plastic characteristics of Di and Am. Also, for Ms and Phn, as shown in Table 3, owing to their higher strength, fracture toughness, and Young's modulus perpendicular to the deconvolution direction, their resistance to deformation is stronger and their EER is greater.

However, for ERP, as illustrated in Figs. 14 and 15, there are some discrepancies with EER since it is only measured in terms of deformation recovery from surface indentations of minerals. For ERP, as illustrated in Figs. 14 and 15, there are some discrepancies with EER since it is only counted in terms of deformation recovery from surface indentations of minerals. Strangely enough, the mean values of ERP for

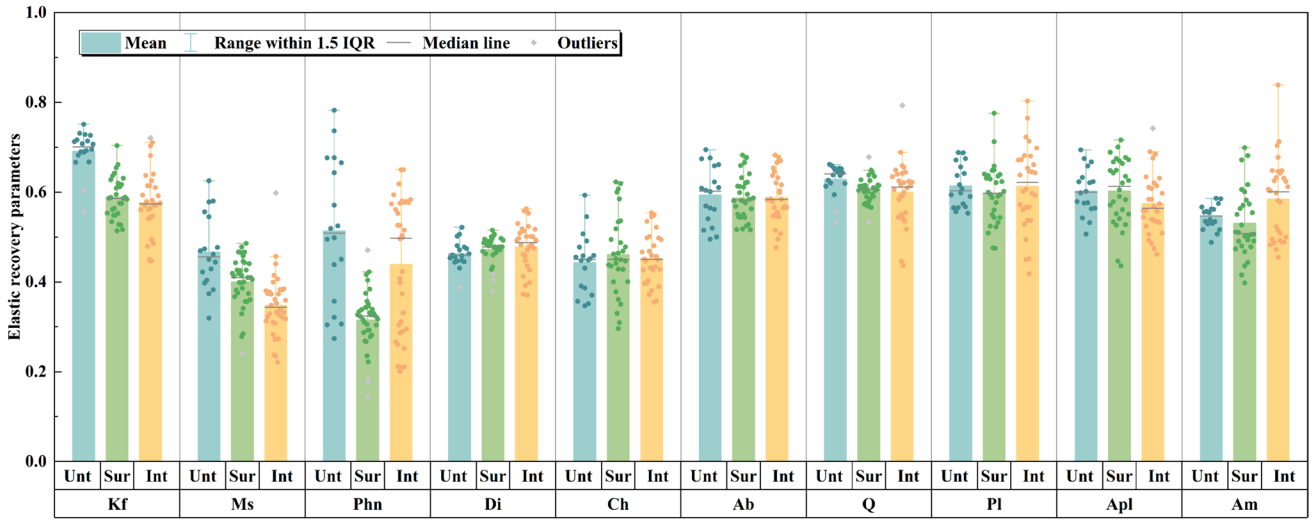


Fig. 15 Elastic recovery parameters of different parts of different minerals under microwave action

soft minerals Ms, Phn, and Ch are commonly larger than the mean values of their EER. On the other hand, the ERP of hard minerals Kf, Ab, Q, Pl, Apl, and Am are commonly smaller than their EER. In addition, the EER of Di is very close to its ERP. Due to these deviations, ERP cannot clearly show the differences in elasticity and plasticity of minerals, i.e., ERP is less effective in assessing the elasticity and plasticity of minerals than EER. Clearly, further exploration is needed to understand the intrinsic mechanism behind this marvelous phenomenon.

Regarding the effect of microwave action, both Yang et al. (2022) and Gao et al. (2023) suggested that microwave irradiation can convert rocks from an elastic state to a plastic state. Considering that the macroscopic physico-mechanical behavior of rocks is determined by the constituent minerals, in the authors' opinion, this phenomenon can also be attributed to the transformation of elastic-plasticity of rock-forming minerals. For example, β-quartz is more susceptible to plastic deformation than α-quartz after high-temperature phase transformation (Evans and Goetze 1979). The elasto-plastic behavior of the minerals in different parts of this research also shows a differential shift. Among them, the ERP and EER of Di, Ab, Q, Pl, and Apl undergo a slight decrease within 5% or even remain unchanged. The ERP and EER decrease more in Kf, Ms, and Phn, with the maximum folding rate reaching more than 10% for Kf and Ms and more than 20% for Phn. Meanwhile, it can be seen from Table 3 that the EER and ERP of Ms and Phn in both directions perpendicular to the solvation direction and parallel to the solvation direction show a significant decrease. However, the ERP and EER for both interior and surface samples of Ch, as well as interior sample of Am, increase inversely. However, there is still a minor decrease in ERP for surface sample of Am. As explained previously, the anomalies of Ch and Am originate from the mineral hardening and the surround and base effects of the internal molten particles, respectively. It can be seen that after microwave irradiation, the irrecoverable plastic deformation of Kf and mica minerals Ms and Phn increases significantly, the elastic recovery of Ch is enhanced, and the plasticity of all other minerals is enhanced to a slight extent.

While elasto-plastic analyses are more focused on the deformation characteristics of samples, brittle–ductile analysis is more indicative of the fracture damage characteristics of minerals. Brittle materials are more susceptible to fracture in a mechanical environment. Therefore, the authors decided to further analyze the link between mineralogy and brittle–ductile response. H/E suggested by Greenwood and Williamson (1966) and Ante et al. (2018) and H/Kc proposed by Lawn and Marshall (1979) were used to evaluate the brittleness of 10 minerals. In addition, the parameters needed for both indexes were included in the parameters of the single loading and unloading test, making it a simple and practical approach. As shown in Table 2, it was found that the minerals Kf, Ab, Q, Pl, Apl, and Am had high brittleness indices (H/E) ranging from 0.10 to 0.16. On the other hand, Ms, Phn, Di, and Ch had relatively weak brittleness with indices ranging from 0.05 to 0.07. Although the brittleness index H/E roughly divides all minerals into two categories, the brittleness indices of the different minerals in the two categories are close to each other. This leads to a less fine-grained division between the brittleness and ductility of the minerals. The brittleness index H/Kc provides a better classification. When comparing the H/Kc of different minerals based on the H/E classification, it is evident that Kf, Q, and Ab are the most brittle, all exceeding 2.40. Q has the highest H/Kc value of 2.86. Following them are Pl, Apl, and Am, with H/Kc values ranging from 1.65 to 2.00. Ms, Phn,

and Ch have the lowest H/Kc values, falling between 0.86 and 0.96. Di has a slightly higher H/Kc value than the three layered silicate minerals, measuring at 1.27. Similar conclusions have been drawn by Nicco et al. (2020) regarding the greater brittleness of quartz, feldspar, and amphibole, as well as the lesser brittleness of mica. In contrast, Tang et al. (2022) concluded that mica is more brittle than quartz and feldspar. This discrepancy arises from Tang et al.'s determination of mineral brittleness solely based on crack lengths of indentations, without quantifying it using parameters.

The brittleness of minerals tested through nanoindentation has also been quantified by some scholars using other parameters. For example, Ante et al. (2018) claimed that brittleness is determined by the duration of the plastic deformation phase. Samples experiencing brittle damage have a shorter plastic deformation period, while samples undergoing ductile damage have a longer plastic deformation period. This phenomenon was also observed in this research. As shown in Table 2, Q, Am, and all four feldspar minerals (Kf, Ab, Pl, and Apl) have a low percentage of plastic energy, ranging from 0.25 to 0.40. Among these, Q and Kf have the lowest plastic energy percentages, both below 0.30. Di and the three layered silicate minerals (Ms, Phn, and Ch) have higher plastic energy percentages in the range of 0.53–0.61, with Di having the lowest plastic energy percentage of 0.53. For Phn and Ms, the brittleness perpendicular to the disintegration direction is slightly higher than that parallel to the disintegration direction (see Table 3). The results of such classification and order are in perfect agreement with the distribution patterns of H/E and H/Kc in the previous section. Others have used the count of pop-in events to quantify the brittleness of minerals, such as Shi et al. (2019) and Espinoza et al. (2022). They generally agree that a higher number of pop-in events represents greater brittleness. This is contrary to the conclusions of the study, where the frequency of pop-in events was instead higher for the tested laminated silicate minerals. The authors argue that this may be attributed to their softer texture and particular laminated structure. In this work, the maximum indentation depth is established to be 1000 nm, which is several orders of magnitude larger than the disintegration pitch. Therefore, when indenting a layered silicate surface, especially in the direction perpendicular to the disintegration direction, the generation and extension of cracks due to penetrating the disintegrated planes can easily occur. This naturally leads to an increase in the number of pop-in events. In the direction parallel to the disintegration direction, it rarely occurs, and this statement can be verified by the morphological characteristics of the curve-loaded segments at different indentation directions shown in Fig. 7b, c. On the other side, the lower number of pop-in events for minerals with high brittleness indexes is attributed to the lower loading rate and the suppression of brittle deformations by the self-contained nature of the indentation method (Kranjc et al. 2016). Elevated temperatures under microwave action typically cause rocks to undergo a transition from brittle to ductile. For instance, Rózanski et al. (2021) found that the "stress–strain" process in sandstones became smoother under temperature cycling. Nevertheless, at the mineral scale, as shown in Table 2 for H/E and H/Kc, the changes in brittleness of different minerals after microwave heating are dissimilar. Specifically, the brittleness of Kf, Di, Ab, Q, and Pl is reduced, while the brittleness of Ch, Apl, and Am is increased. It is worth noting that Apl only exhibits an increase in brittleness in the surface sample. This may be attributed to the decomposition of its internal carbonates at high temperatures, generating expansion air pressure from the inside out. This leads to the closure of some of the microfractures in the sample and an increase in the hardness of Apl, which is more pronounced in its surface hardness. The brittleness variations of Ms and Phn also differed, with Ms showing a decrease in brittleness for the different indentation directions. However, the brittleness of Phn increased in the direction parallel to the disintegration direction. The brittleness of the surface sample decreases, while the interior sample increases in the direction perpendicular to the solvation direction. To the authors, this can be attributed to the inclusion of a higher content of quartz impurities inside Phn, which leads to its lower heating rate and poorer weakening effect. Alternatively, it can be attributed to the good thermal expansion properties of quartz, which allows mica minerals in its vicinity to be compacted.

3.6 Melt Composition and Micro-mechanical Properties

The final consequence of microwave radiation for some rocks is melting rather than crushing, usually due to the presence of microwave-sensitive minerals in the rock. For example, in the Anshan gabbro examined by Lu et al. (2020a), pyroxene is a good microwave absorber. However, the specific composition and content of the melted rocks or minerals are still unknown, and research on the mechanical properties of the various components of the melt has not been conducted. Therefore, the authors will conduct relevant studies. The response of Ch, Apl, and Am to microwave irradiation at a power of 2 kW is sensitive, with Am and Ch experiencing melt destruction at 82 s and 110 s, respectively, and their melt morphologies are shown in Fig. 16a, b. Table 5 records the different components of Ch, Am, and their melts after microwave irradiation with the corresponding contents tested using XRD. Following microwave irradiation, the Ch fraction exhibited the disappearance of chlorite and minor quantities of albite, while plagioclase experienced a substantial reduction. In the new components, alongside substantial quantities of amorphous melt, there were small amounts of



(a) Ch (b) Am (c) Melt samples

Fig. 16 Melts morphology of Ch and Am

dicalcium silicate, mica, and magnetite, as well as a notable increase in quartz. In the authors' opinion, there is little doubt that chlorite is a reliable microwave absorber, and that the isomorphous substitution of chlorite at high melting temperatures contributes to the formation of mica. In addition, the roasting of chlorite and feldspar minerals at high temperatures also facilitates other redox reactions, including the release of water molecules and the creation of products such as CaO, SiO_2, and Fe_3O_4. These reactions directly lead to an increase in the presence of quartz and magnetite. Subsequently, CaO and SiO_2 continued to react through chemosynthesis under melting conditions to produce dicalcium silicate. A similar pattern exists for the variation in the melt composition of Am. As shown in Table 5, after microwave exposure, the amphibole content decreases from 72% to as little as 7.9% after melting. In addition, a small amount of biotite disappears and the plagioclase content also decreases. In the nascent mineral component, apart from a significant amount of amorphous melt, there is also a minor proportion of pyroxene, accompanied by a significant increase in quartz. This indicates that amphibole is a good microwave absorber and undergoes redox reactions in which it loses hydroxide under the effect of the molten hydrothermal fluid, resulting in its conversion to pyroxene. At the same time, in the high-temperature chemical reaction between amphibole and plagioclase, water molecules escape and products such as SiO2 appear. SiO2 is not further bound, resulting in a significant increase in quartz content. In addition, the feldspar minerals in the melts of both Ch and Am are reduced, confirming the prediction of Zeng et al. (2019) about the location of the origin of the melt cavities. This location is in zones of crack-dense feldspar minerals near the microwave-sensitive mineral-rich zones.

In this research, a combination of XRD analysis, lattice nanoindentation technique (3 × 3 matrix), and the K-means clustering algorithm (Hartigan and Wong 1979) was used to determine the mean values of mechanical parameters for separated components from the collection of nanoindentation test results on melt samples (See Fig. 16c). For detailed information on the principles and process of the K-means clustering algorithm, the authors refer the reader to the work of Lu et al. (2022). Only the decomposition framework of the algorithm is provided here.

1. There exists a dataset of size n. Set $I = 1$ and select k initial clustering centres $Z_j(I)$, j = 1, 2, 3,…, k;

2. Calculate the distance from each data object to the cluster centre as $D(x_i,Z_j(I))$, and if it satisfies $D(x_i,Z_k(I)) = \min\{D(x_i,Z_j(I)), j = 1, 2, 3,\ldots, n,\}$ then $x_i \in w_k$;

3. Calculate the squared error and the criterion function:

$$J_c(I) = \sum_{j=1}^{k} \sum_{k=1}^{n_j} \left\| x_k^{(j)} - Z_j(I) \right\|^2 \tag{15}$$

4. Judgement: If $\left|J_c(I) - J_c(I-1)\right| < \xi$, the algorithm ends and the parameter cluster is divided into k clusters.

Table 5 Composition and content of Ch and Am minerals and corresponding melts

Mineral type	Component a (Content/%)	Component b (Content/%)	Component c (Content/%)	Component d (Content/%)	Component e (Content/%)	Component f (Content/%)
Ch	Chlorite (45.0%)	Plagioclase (46.0%)	Albite (5.0%)	Quartz (3.0%)	Epidote (1.0%)	–
Melt of Ch	Mica (4.1%)	Plagioclase (10.6%)	Dicalcium silicate (6.2%)	Quartz (10.2%)	Magnetite(3.0%)	Amorphous (66.0%)
Am	Amphibole (72.0%)	Plagioclase (20.0%)	Biotite (4.0%)	Quartz (2.0%)	Carnegieite (1.0%)	Sphene (1.0%)
Melt of Am	Amphibole (7.9%)	Plagioclase (16.8%)	Pyroxene(5.1%)	Quartz (24.6%)	Amorphous (45.6%)	–

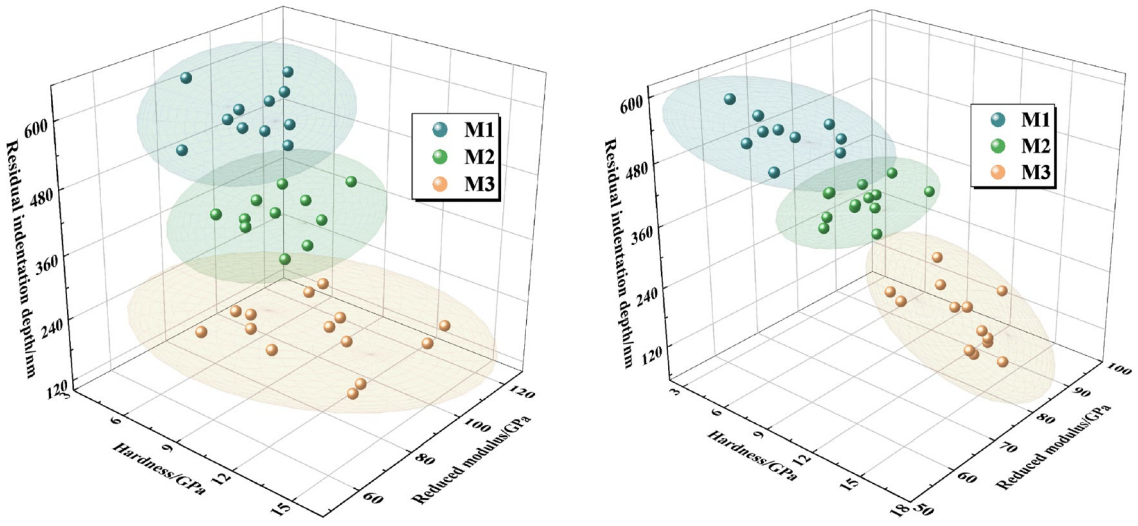


Fig. 17 Classification results of K-means clustering algorithm for Ch and Am melts based on nanoindentation data: (**a**) Ch and (**b**) Am

Otherwise, set $I = I + 1$, calculate k new cluster centres, $Z_j(I) = \frac{1}{n}\sum_{i=1}^{n_j} x_i^{(j)}$, and return to step 2.

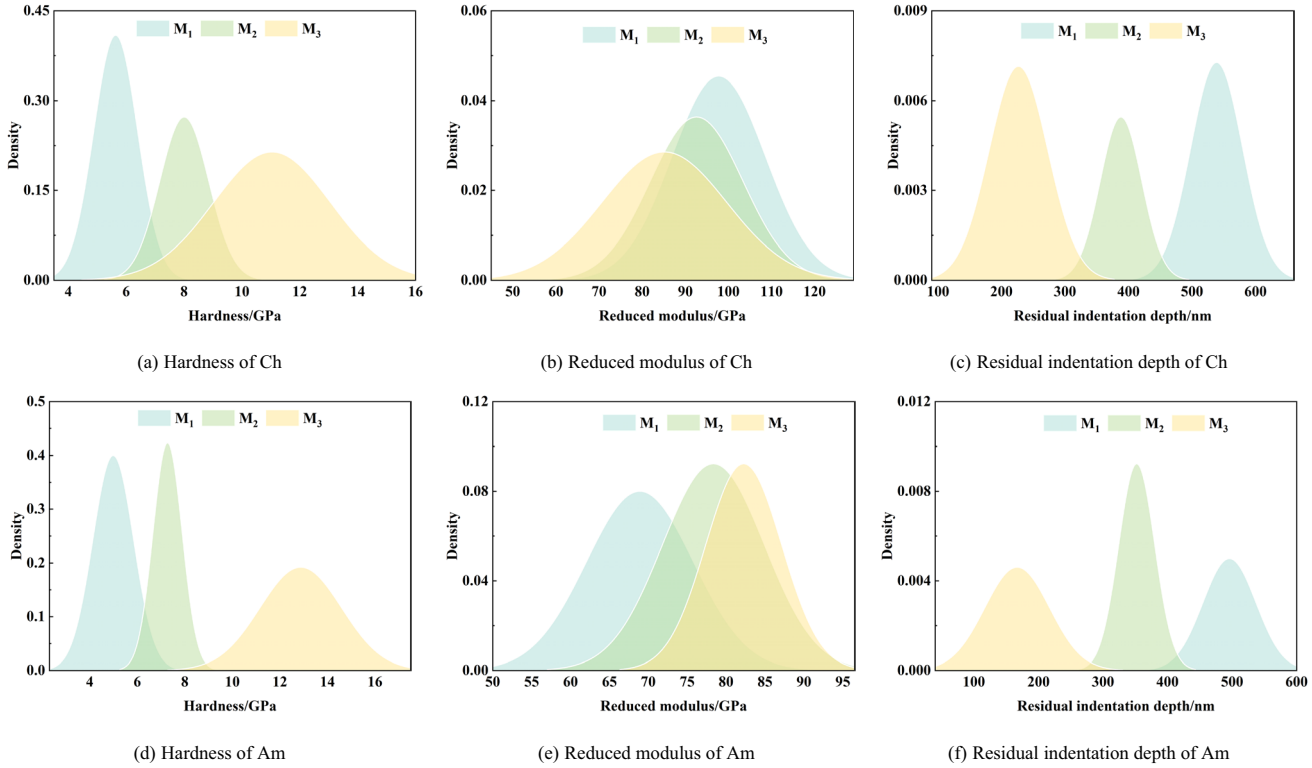
According to three indicators of Ch and Am's melts: hardness, reduced modulus, and residual indentation depth, the mineral phases were classified using a K-means clustering algorithm. As can be seen in Fig. 17, the mineral phases of both Ch and Am melts were classified into three grades: M1, M2, and M3, based on their strength. The corresponding probability density functions for hardness, reduced modulus, and residual depth are shown in Fig. 18. The values of the horizontal coordinates corresponding to the peaks are the clustering center values of the mineral phases and also the mean values of the mineral phases, which were uniformly recorded in Table 6. In the melt of Ch, mineral phases M1, M2, and M3 have hardnesses of 5.64 GPa, 8.00 GPa, and 11.04 GPa, respectively, reduced modulus of 97.75 GPa, 92.61 GPa, and 85.12 GPa, respectively, and residual depths of 538.71 mm, 388.06 mm, and 226.64 mm, respectively. Combined with the results of XRD analysis, it can be concluded that on a microscopic scale, taking into account the variability of different components' mechanical properties, the mineral components in the melt of Ch can be categorized into three groups: high-strength minerals, medium-strength minerals, and low-strength minerals. Among these, the high-strength minerals consist mainly of amorphous and quartz, the medium-strength minerals include plagioclase and magnetite, and the low-strength minerals are primarily dicalcium silicate and mica. On the other hand, the hardest mineral phase, M3, has the lowest reduced modulus, which again confirms the fact of poor correlation between hardness and reduced modulus. For the melt of Am, the mineral phases M1, M2, and M3 have.

4 Discussion

4.1 Indentation Morphology—Micro-mechanical Parameters Connection

Indentation morphological features provide basic information about material strength and its scale-related deformation properties (Lips and Sack 1936). Typical indentation matrix topographies were chosen to compare the variability of the topographic features of the indentation points at different sites of the 10 minerals before and after microwave action, as shown in Fig. 5. As far as the residual indentation morphology is concerned, the indentation areas of Q, Kf, Ab, Pl, Apl, and Am are smaller, while those of Ms, Phn, Di, and Ch are larger. This difference in size is related to the elastic properties of different minerals. Since all minerals have the same maximum indentation depth, their maximum indentation area is also identical. Therefore, minerals with a high percentage of elastic recovery have a smaller residual indentation area. This finding is consistent with the pattern presented by minerals' ERP and EER in Table 2.

Although cracks should be prevented in nanoindentation tests, differences in the mechanical properties and structure of different minerals make it impossible to avoid the appearance of cracks. Three typical crack patterns can be observed in indentation, namely lateral, radial, and shear cracks, as summarized by Liu et al. (2022). The pattern of crack extension is entirely distinct for different minerals, and the authors consider that it can be linked to the mechanical properties and microstructural features of minerals. For several minerals, such as Q, Kf, Ab, Pl,



(a) Hardness of Ch (b) Reduced modulus of Ch (c) Residual indentation depth of Ch

(d) Hardness of Am (e) Reduced modulus of Am (f) Residual indentation depth of Am

Fig. 18 Probability density functions of the mechanical properties of the melts of Ch and Am

Table 6 Analysis results of K-mean clustering of nanoindentation data for melts of Ch and Am

Mineral type	Number of clusters	Number of cases	Final clustering center for hardness/GPa	Final clustering centers for Young's modulus/GPa	Final clustering center for residual depth/nm
Ch	Cluster 1	11	5.64	97.75	538.71
	Cluster 2	11	8.00	92.61	388.06
	Cluster 3	14	11.04	85.12	226.64
Am	Cluster 1	10	4.98	68.91	495.84
	Cluster 2	13	7.27	78.37	352.19
	Cluster 3	13	12.88	82.26	167.00

Apl, and Am, which have high brittleness and hardness, the phenomenon of chip spalling occurs near the indentation. This is because highly brittle solids are prone to the formation of radial and lateral cracks in the deformation zone during the indentation test (Lawn et al. 1980; Lawn and Swain 1975). When the lateral cracks connect with the radial cracks, it leads to the chipping phenomenon in brittle minerals (Liu et al. 2022). However, for the mineral Di, which has high hardness and moderate brittleness, the residual indentation morphology is clearer and more complete, with fewer radial cracks. For the minerals Ms, Phn, and Ch, which have low brittleness and hardness, larger areas of flake spalling are present near the surface residual indentation of Ch and Phn (perpendicular to the solvation direction) due to their special laminate structure and low interlayer adhesion. In contrast, the indentation morphology of Ms perpendicular to the disintegration direction is consistent with the shape of the indenter, but the presence of discontinuous shear bands around the indentation can also be obviously seen. The formation of these shear cracks has mainly been attributed to dislocations and slips under indenter pressure due to the disordered silicate lamellar stacking (Liu et al. 2022), as well as severe surface deformation when the indenter is in contact with the amorphous phase (Chen et al. 1994; Kim et al. 2002). It is worth noting that both the formation of shear brands and the large flake spalling near the indentation contribute to the anomalous slope variations of the curves

during the loading phase (Liu et al. 2022). The authors suggest that these are the main reasons why Ch, as well as Ms and Phn perpendicular to the deconvolution direction, have a higher frequency of pop-in events. On the other hand, when indenting parallel to the disintegration direction, there is no longer significant flake spalling and shear bands, but rather cracks that extend along the disintegration direction. It can be seen that for layered silicate minerals, when the angle between the indentation direction and the disintegration direction is different, the crack extension mechanism around the indentation is distinct, leading to different measured mechanical properties.

After microwave irradiation, the morphology of the mineral surfaces' indentation changed significantly. This change mainly included the phenomenon of dropping blocks in the affected area, the length and distribution of cracks near the indentation, and the size of the residual indentation. As shown in Table 2, minerals Q, Kf, Ab, and Pl exhibited similar development characteristics in their mechanical properties after microwave heating. Specifically, there was a slight decrease in hardness, fracture toughness, two elasticity indexes (EER, ERP), and two brittleness indexes (H/E, H/Kc). In conjunction with Fig. 5a, f, g, h, the decrease in the two elasticity metrics resulted in slightly larger surface indentation areas for both surface and internal mineral samples. It should be mentioned that some of the indentation points show a larger area of irregular triangles when not heated by microwave. This is ascribed to the block flaking near the indentation of the highly brittle minerals, which makes them appear larger in area. On the other hand, the decrease in the two brittleness indices weakened the frequency of block dropout. In addition, the decrease in hardness and fracture toughness increased the length of cracks induced during indenter press-in, particularly the length of radial cracks. For Apl, which is also a feldspar mineral, the mechanical properties and surface indentation morphology of the samples in different parts after microwave action were unlike those of other feldspar minerals. When heated for 45 s, Apl was damaged by the thermal expansion air pressure generated by the decomposition of internal carbonates and disintegrated violently. At the same time, the thermal expansion allowed for the strengthening of the mechanical properties of Apl. This is demonstrated in Table 2, where the hardness, fracture toughness, and H/Kc of Apl's surface sample increased, while other mechanical parameters remained unchanged. The strengthening effect on the interior sample was slightly less pronounced, with a minor increase in hardness and fracture toughness, and a slight decrease in other parameters. This could be attributed to the growth of bulky holes within its interior sample. The corresponding phenomenon can be found in Fig. 5i. For example, the increase in hardness and fracture toughness led to a decrease in crack length. Meanwhile, the slight changes in the elasticity and brittleness indexes caused negligible changes in the indentation area and the number of chips, which may be related to the fact that Apl is destroyed after a relatively short period of microwave irradiation. Am weakens the most after microwave irradiation. After 82 s of microwave irradiation, Am melted, and the hardness and fracture toughness of the surface sample decreased significantly. The elasticity parameter remained almost unchanged, but the brittleness index was strengthened. The hardness and fracture toughness of the interior sample are much larger than those of the surface sample, and the elasticity parameter rises markedly. The brittleness indexes are also strengthened, similar to the surface sample. Corresponding indentation morphology features include a smaller residual indentation area and more severe dropout. Curiously, the interior sample has greater hardness and fracture toughness but exhibits more pronounced radial cracking. This reaffirms the fact that Am's interior sample actually has a lower fracture toughness, and that it is the surround and base effects of the melt that bias their test results.

Another mineral that melts after 110 s of microwave heating is Ch. Ch is an all-round strengthened mineral with not only increased hardness and fracture toughness but also increased elasticity and brittleness indices. A significantly reduced indentation area can be found in Fig. 5e, and although the increase in brittleness has resulted in small-sized dropouts and complex cracking patterns near the indentation point of Ch, none of the cracks in Ch have been able to extend further because of its enhanced hardness and fracture toughness. For the other two layered silicate minerals, Ms and Phn, there is a decrease in hardness, fracture toughness, elasticity indices, and brittleness indices in the direction perpendicular to the disintegration direction. This pattern is similar to the evolutionary pattern observed in Kf, Q, Ab, and Pl, including the indentation morphology. However, parallel to the disintegration direction, the hardness and fracture toughness of Ms and Phn increase, while the elasticity index decreases. The difference lies in the fact that the brittleness index decreases for Ms and increases for Phn. From Fig. 5b4–6, c4–6, it can be observed that the energy imbalance between the hemispherical plastic zone formed under the indenter and the outer elastic zone is preferentially released through the soft disintegration surface, due to the presence of disintegration interstices. This results in no chip spalling around the indentation of both the microwave-treated brittleness-reduced Ms and the brittleness-elevated Phn. Instead, it leads to the creation of nascent fissures along the direction of disintegration on the surface of Phn's surface sample and interior sample surfaces. This phenomenon is absent on the surface of the Ms. In other words, the presence of weak deconvolution surfaces would allow the release of brittle fracture energy through nascent fissures on the adjacent deconvolution surfaces to the indenter. Of course, the decrease in the elasticity index also causes an

increase in the area of residual indentation to occur in both cases. As far as Di is concerned, its hardness, fracture toughness, and brittleness indexes declined slightly after microwave irradiation, while the elasticity index remained almost unchanged. Therefore, as shown in Fig. 5d, the indentation area of Di remains almost constant after microwave exposure. The lower brittleness allowed the integrity of the Di indentation to remain intact, with only isolated small pieces of chip spalling. However, the lower fracture toughness and denser porosity allowed connected cracks to form between the indentations on the right side of Di's interior sample.

4.2 Dielectric Properties—Heating Rate—Micro-mechanical Parameters Connection

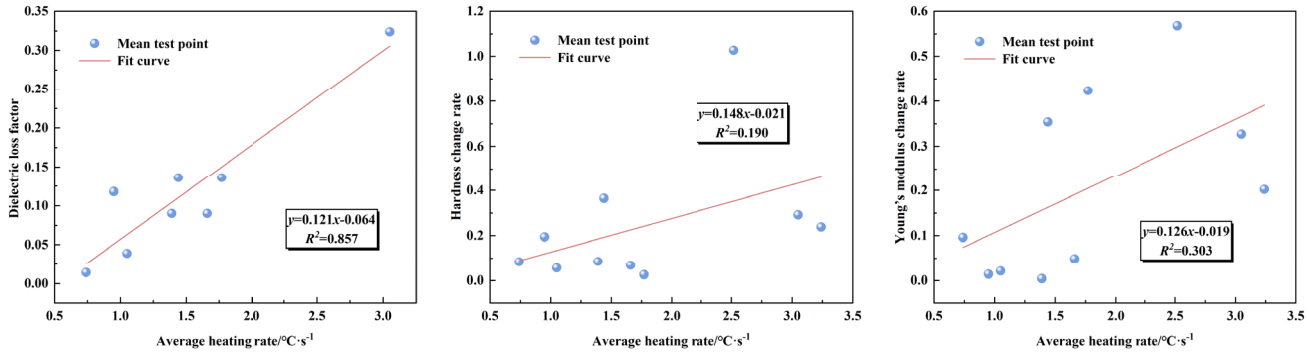
Lu et al. (2020a) investigated the relationship between the dielectric properties, microwave heating effect, and uniaxial compressive strength of rocks. The pattern discovered was that the stronger the dielectric properties of rocks, the faster the heating rate, the greater the decline in uniaxial compressive strength of rocks, and the better the fracturing effect. According to the authors, the effect of microwave heating and the weakening of rocks or minerals depends on two types of factors. One factor is the environment, which includes the electric field strength, microwave frequency, and irradiation conditions. These factors were all consistently set up in this study. The other factor is the sample itself, which includes the density, specific heat capacity, and loss factor of each mineral. Considering that there is little difference in the density and specific heat capacity of each type of mineral, the heating rate of minerals is mainly related to the loss factor. A similar conclusion was reached by Zhao et al. (2020). By examining the dielectric properties tested by other authors (Nelson et al. 1989; Zheng et al. 2020) at 2.45 Hz and room temperature, the dielectric loss factors of the corresponding minerals (excluding Ch and Apl) and the heating rates tested in this study are summarized in Table 2. Considering the variability in the change rate of mechanical properties between surface and interior mineral samples under microwave irradiation, coupled with the fact that interior samples are more likely to be affected by randomly distributed melt, the absolute values of the change rate of hardness or Young's modulus of minerals' surface samples were used to quantify the change rate of mechanical parameters of minerals on the nanoscale. The statistical results are presented in Table 2. This also enables the linkage of the change rate of mechanical parameters on the surface with the results of the monitoring of the surface temperature.

As shown in Fig. 19a, the relationship between the dielectric loss factor and the heating rate exhibits a well-fitted linear relationship with a linear correlation coefficient R^2 of 0.857. The small fitting error can be attributed to the difference in the testing techniques of the dielectric properties (Zheng et al. 2020) or fluctuations in the dielectric properties of the minerals during temperature change (Campbell and Ulrichs 1969; Zheng 2018). However, as shown in Fig. 19b, c, the coordinate point distributions of the change rate of hardness or Young's modulus versus the mineral's heating rate are more discrete, with poorer linear fitting relationships, and linear correlation coefficients of R^2 of only 0.303 and 0.190. Specifically, among them, the coordinate points of the three layered silicate minerals (Ms, Phn, and Ch) deviate the most from the fitted curves, causing a poorer linear fitting relationship, which is most likely driven by their particular structures. Undoubtedly, the change rate of hardness or Young's modulus reflects and quantifies the real effect that the same microwave irradiation conditions have on the mechanical properties of different rock-forming minerals. However, Fig. 18b, c shows its weak correlation with the microwave sensitivity (heating rate or dielectric loss factor) of minerals. In other words, the strength reduction and rupture patterns of hard rocks cannot be accurately predicted solely based on the thermo-physical or electrical properties of the rock-forming minerals. Instead, the focus should be on the variation characteristics of the mechanical properties of the minerals after exposure to microwave radiation.

5 Conclusions

This research tested the micro-mechanical properties of 10 major granite-forming minerals. A series of mechanical parameters, such as hardness, Young's modulus, and fracture toughness, were used to quantify the effects of microwave irradiation on the surface and interior micromechanical properties of different minerals. The study also examined mineral fractions and corresponding mechanical properties of melts of molten minerals. In addition, the effects of microwave irradiation on the surface topography and indentation morphology of the mineral samples were observed with the aid of a microscope. The specific conclusions are as follows:

1. Ten minerals are roughly divided into three categories by hardness: Di, Q, and Am are high-hardness minerals, with hardnesses above 11 GPa. All three of them have high Young's modulus and fracture toughness, as well as low creep depths and discrete curve distributions. The four feldspar minerals, Kf, Ab, Pl, and Apl, are medium-hardness minerals, with hardnesses in the range of 6–11 GPa. Their other micro-mechanical parameters are also medium. Three layered silicate minerals, Ms, Phn, and Ch are low-hardness minerals, and their micro-mechanical parameters present a completely opposite pattern to that of high-hardness minerals. Therefore, when the content of high-hardness minerals is high in granite,



(a) Relationship between dielectric loss factor and average heating rate

(b) Relationship between hardness change rate and average heating rate

(c) Relationship between Young's modulus change rate and average heating rate

Fig. 19 Fitting of dielectric loss factor, hardness change rate and Young's modulus change rate to average heating rates of the 10 minerals

mechanical cutters are more likely to be worn out during excavation and mining is less efficient.

2. The EER in the two elasticity parameters and the H/Kc in the two brittleness parameters are better differentiated. There is a link between the classification of minerals based on elastic–plastic and brittle–ductile properties and the classification based on parameters such as hardness and Young's modulus. Specifically, high- and medium-hardness minerals have better elasticity (EER ≥ 0.59) and brittleness (H/Kc ≥ 1.65), while low-hardness minerals have locally better plasticity (EER ≤ 0.46) and ductility (H/Kc ≤ 0.96). Di, however, is a counter-example. Despite being a high-hardness mineral (H = 11.42 GPa), it exhibits better plasticity and ductility (EER = 0.47, H/Kc = 1.27).
3. Ch, Apl, and Am are all microwave-sensitive minerals, which melt or disintegrate at 110 s, 45 s, and 82 s, respectively, with a heating rate of > 2.5 °C/s. Of these, the surface sample of Am exhibits a high discount rate of mechanical parameters, while the interior sample exhibits a low discount rate of mechanical parameters due to the surround and base effects of the melt. Ch and Apl are both strengthened, but through different mechanisms. Ch is strengthened by high-temperature dehydration, which causes all-round strengthening due to the contraction of the solvent spacing. Apl, on the other hand, is locally strengthened (mainly on the surface) caused by the inner carbonate decomposition at high temperature, resulting in expansion air pressure. Therefore, microwave pretreatment is more suited to be applied to granite or other hard rocks with high Am content, which are weakened better. This results in slower wear of mechanical tools and more efficient excavation.
4. Q, Di, Kf, Ab, and Pl are poor microwave absorbers, with a heating rate of < 2 °C/s. Their micro-mechanical parameters have a low discount rate, and the interior samples have a slightly better discount rate. Therefore, after microwave irradiation, the weakening effect of rocks containing more of these minerals is not evident, and the excavation efficiency improvement is minor. The microwave sensitivities of Ms and Phn are also poor, but their mechanical parameter change rates are abnormal, reaching up to 42.3%. This abnormality may be attributed to their laminar disintegration structure. In addition, the mic-mechanical parameters exhibit inconsistent changes under microwave effects in various indentation directions. Notably, the direction parallel to the laminations is more suitable to be utilized as the forward direction for cutting and drilling where the mechanical properties of Ms and Phn are poorer.
5. After microwave irradiation, the mineral samples all showed surface deterioration, including an increase in holes, cracks, and melts of different sizes. The deterioration tended to be more pronounced in the interior samples. Among them, the microwave-sensitive minerals Apl and Am showed the most significant deterioration effects. The variability in indentation morphology depends mainly on the fracture toughness, elasticity, and brittleness parameters of minerals. For example, lower fracture toughness, elasticity, and brittleness parameters usually lead to longer radial cracks, larger residual indentation areas, and fewer chips.
6. The linear fit of the heating rate and dielectric loss factor of minerals is excellent, with a linear correlation coefficient R^2 of 0.857. However, the coordinates of hardness and Young's modulus change rate versus mineral heating rate are discretely distributed with poor linear fit, and the linear correlation coefficients, R^2, are only 0.303 and 0.190, respectively. This shows that the heating rate and dielectric properties of minerals are not suitable to be used to quantify the microwave response of their mechanical parameters. In other words, under

microwave action, the strength reduction, cracking and destruction of hard rock cannot be predicted by merely measuring the thermophysical and electrical properties of minerals, but more attention should be paid to the variation patterns of minerals' mechanical properties.

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Data Availability The related data used to support the study of this article are available from the corresponding author upon reasonable request.

Declarations

Conflict of Interest All authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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